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Qi et al.

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(54) **SYSTEMS AND METHODS FOR
SUPPRESSING SOUND LEAKAGE**

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Feb. 9, 2021, now Pat. No. 11,582,565, which is a
(Continued)

(30) **Foreign Application Priority Data**

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Aug. 24, 2018 (CN) 201810975515.1

(51) **Int. Cl.**
H04R 1/04 (2006.01)
G10K 9/13 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H04R 25/505** (2013.01); **G10K 9/13**
(2013.01); **G10K 9/22** (2013.01); **G10K 11/175**
(2013.01);
(Continued)

(58) **Field of Classification Search**

CPC G10K 11/175; G10K 11/178; H04R 1/04;
H04R 1/1086; H04R 1/2811; H04R
2460/13

(Continued)

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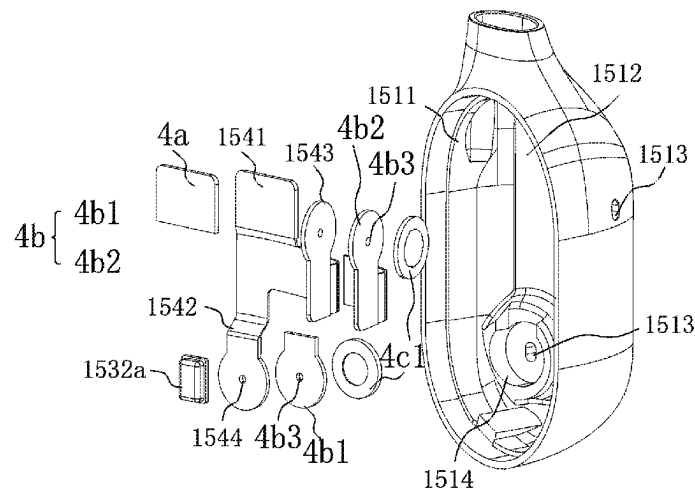
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(57) **ABSTRACT**

A speaker comprises a housing, a transducer residing inside the housing, and at least one sound guiding hole located on the housing. The transducer generates vibrations. The vibrations produce a sound wave inside the housing and cause a leaked sound wave spreading outside the housing from a portion of the housing. The at least one sound guiding hole guides the sound wave inside the housing through the at least one sound guiding hole to an outside of the housing. The guided sound wave interferes with the leaked sound wave in a target region. The interference at a specific frequency relates to a distance between the at least one sound guiding hole and the portion of the housing.

16 Claims, 20 Drawing Sheets



Related U.S. Application Data

continuation-in-part of application No. 17/074,762, filed on Oct. 20, 2020, now Pat. No. 11,197,106, which is a continuation-in-part of application No. 16/813,915, filed on Mar. 10, 2020, now Pat. No. 10,848,878, which is a continuation of application No. 16/419,049, filed on May 22, 2019, now Pat. No. 10,616,696, which is a continuation of application No. 16/180,020, filed on Nov. 5, 2018, now Pat. No. 10,334,372, which is a continuation of application No. 15/650,909, filed on Jul. 16, 2017, now Pat. No. 10,149,071, which is a continuation of application No. 15/109,831, filed as application No. PCT/CN2014/094065 on Dec. 17, 2014, now Pat. No. 9,729,978, said application No. 17/171,987 is a continuation-in-part of application No. 16/950,876, filed on Nov. 17, 2020, now Pat. No. 12,105,357, which is a continuation of application No. PCT/CN2019/102394, filed on Aug. 24, 2019.

(51) **Int. Cl.**

G10K 9/22 (2006.01)
G10K 11/175 (2006.01)
G10K 11/178 (2006.01)
G10K 11/26 (2006.01)
H04R 1/08 (2006.01)
H04R 1/28 (2006.01)
H04R 9/06 (2006.01)
H04R 25/00 (2006.01)
H04R 17/00 (2006.01)

(52) **U.S. Cl.**

CPC **G10K 11/178** (2013.01); **G10K 11/26** (2013.01); **H04R 1/04** (2013.01); **H04R 1/086** (2013.01); **H04R 1/2811** (2013.01); **H04R 9/066** (2013.01); **H04R 25/405** (2013.01); **G10K 2210/3216** (2013.01); **H04R 1/2876** (2013.01); **H04R 17/00** (2013.01); **H04R 2460/13** (2013.01)

(58) **Field of Classification Search**

USPC 381/151
 See application file for complete search history.

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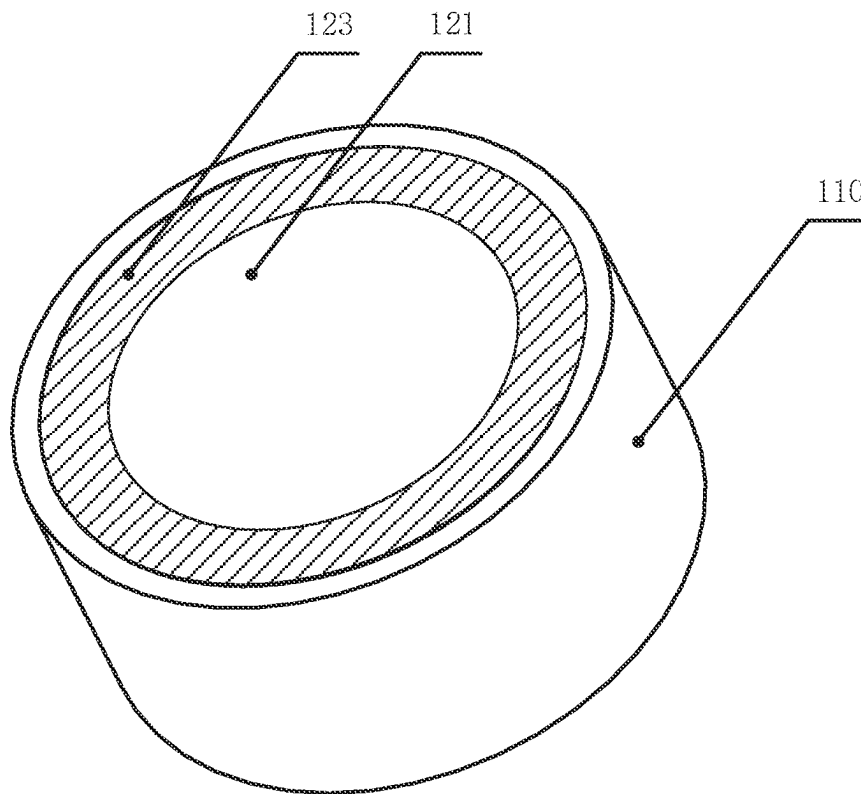


FIG. 1A

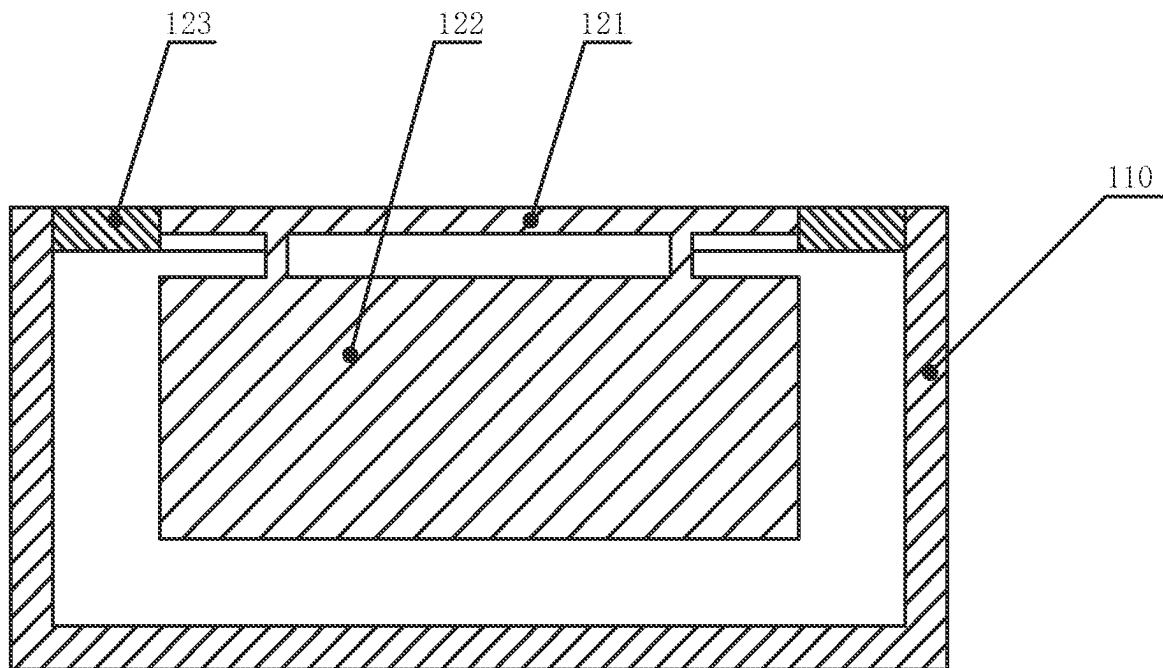


FIG. 1B

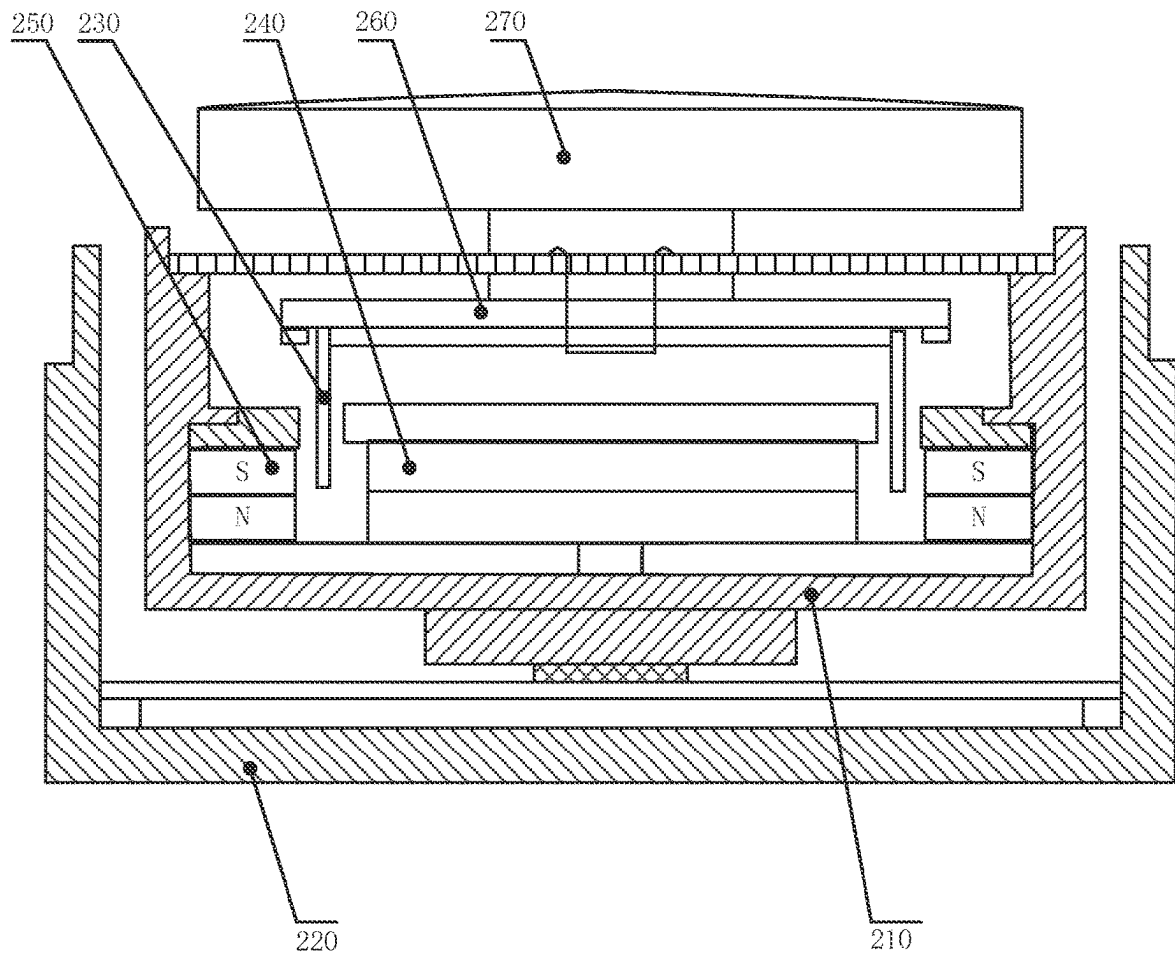


FIG. 2

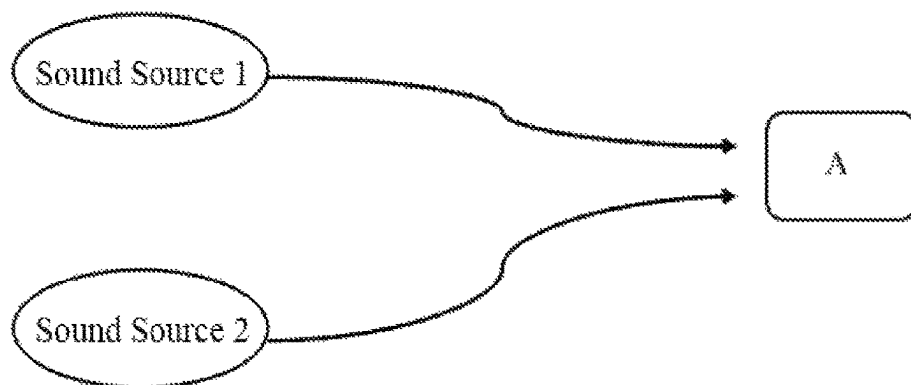


FIG. 3

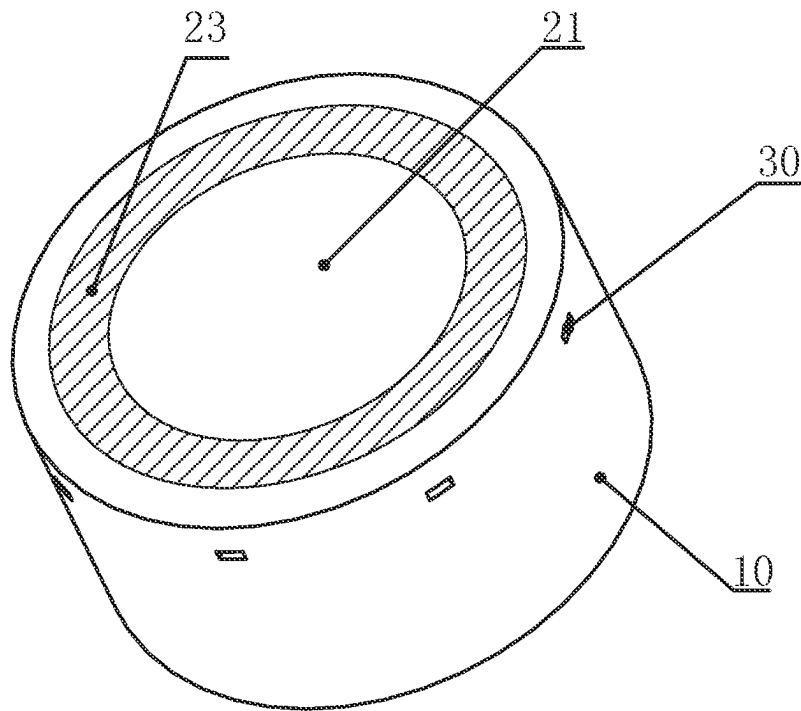


FIG. 4A

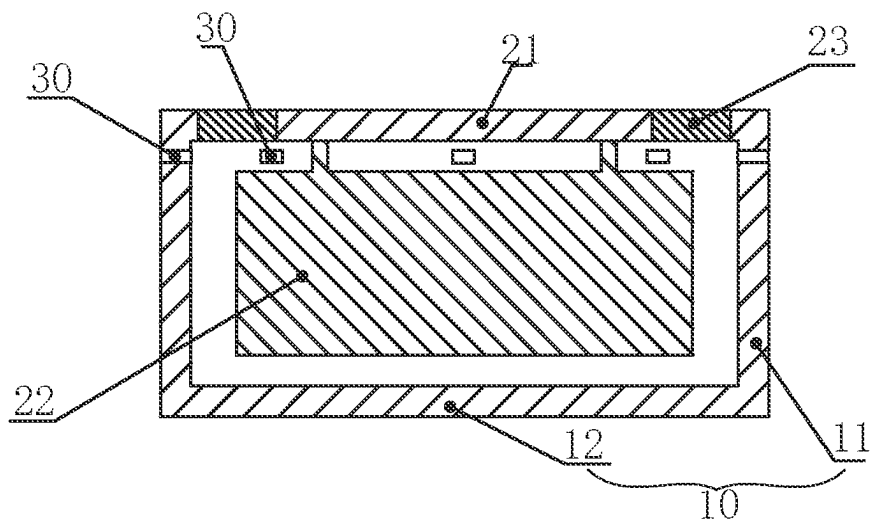


FIG. 4B

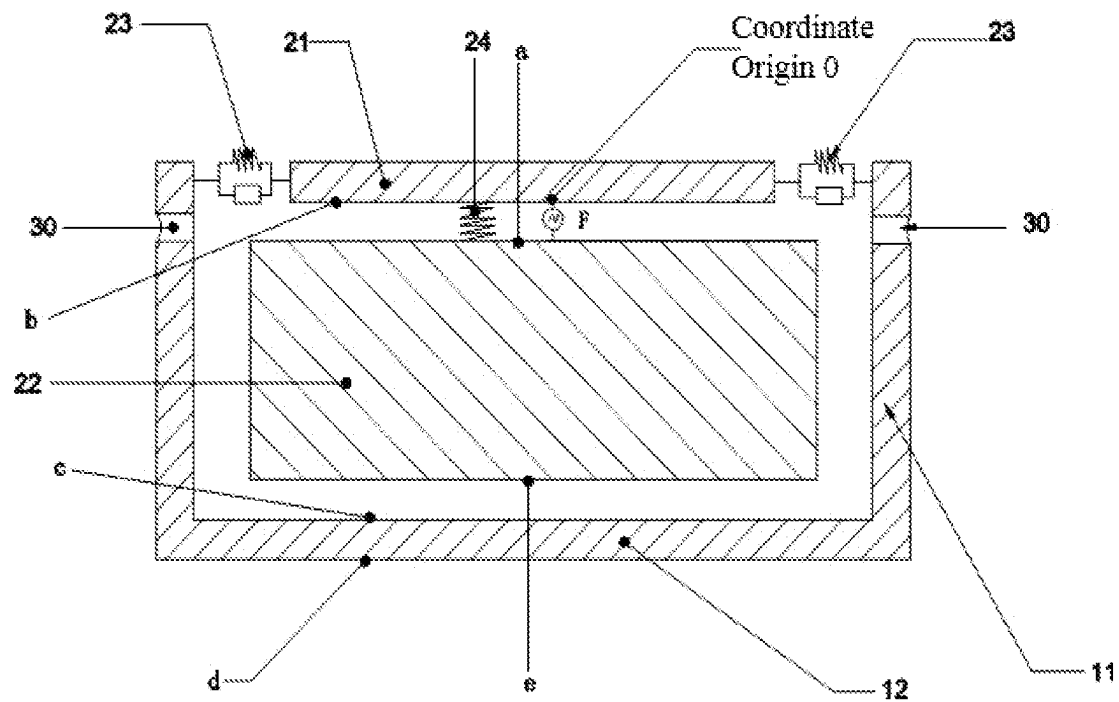


FIG. 4C

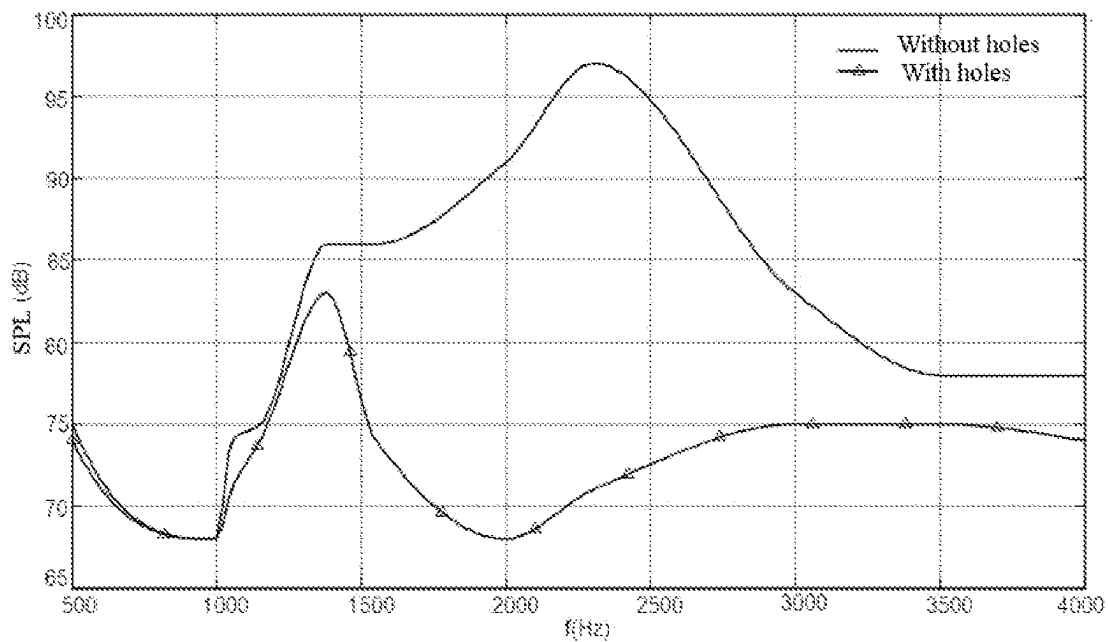


FIG. 4D

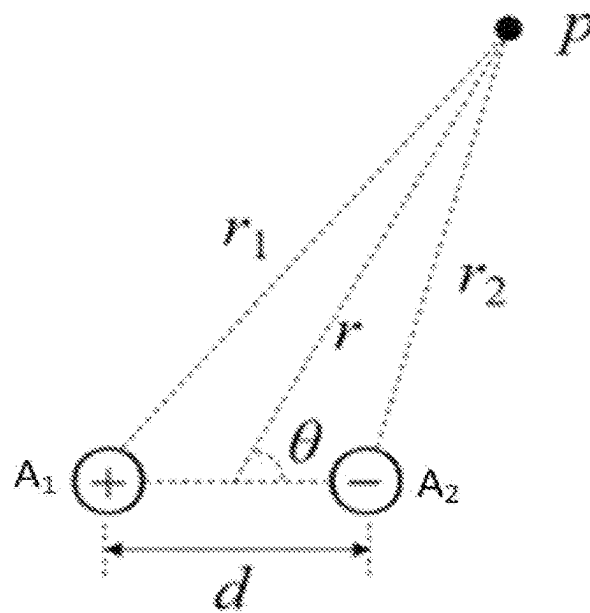


FIG. 4E

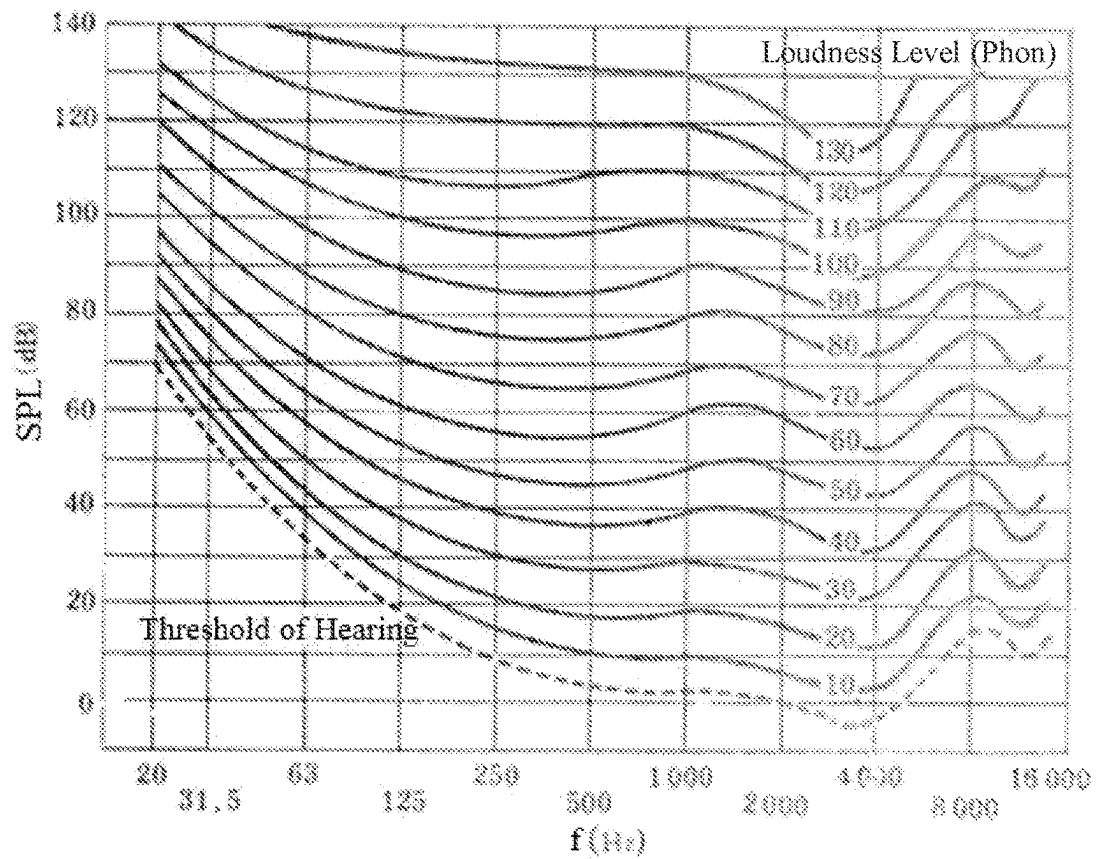


FIG. 5

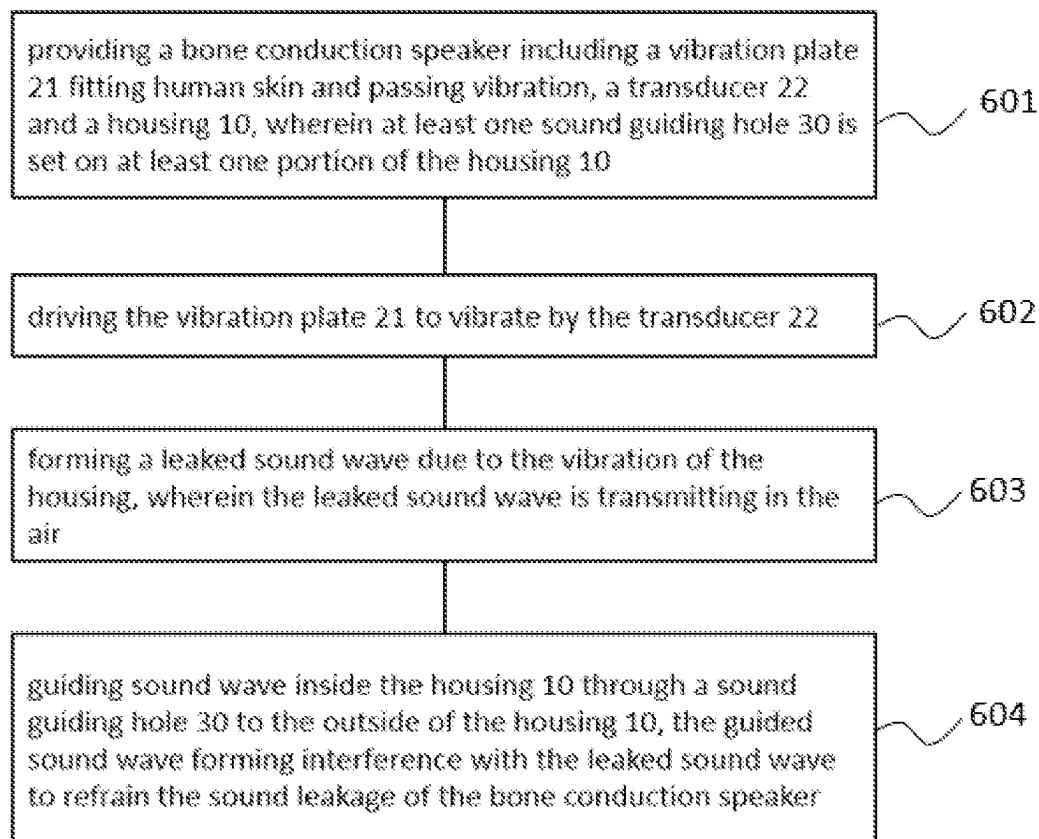


FIG. 6

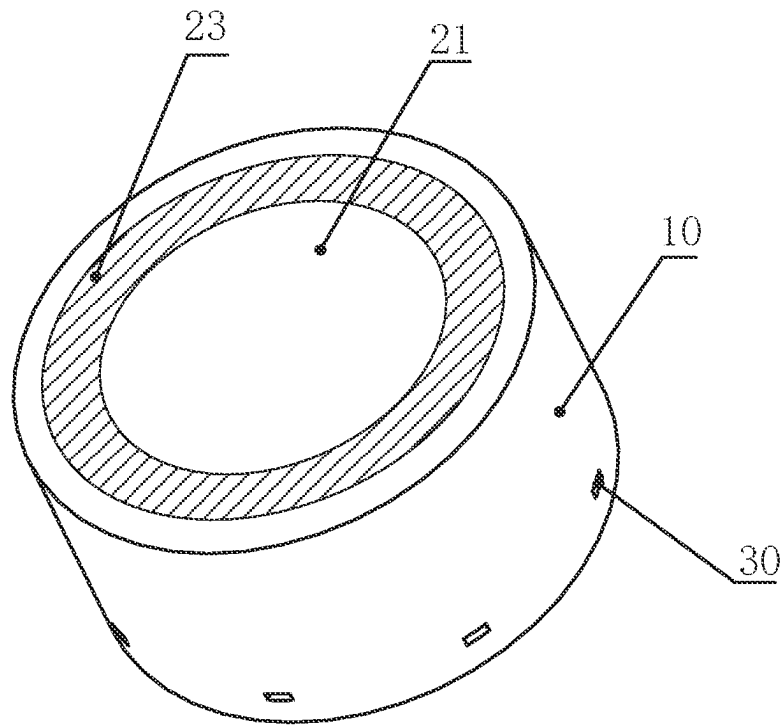


FIG. 7A

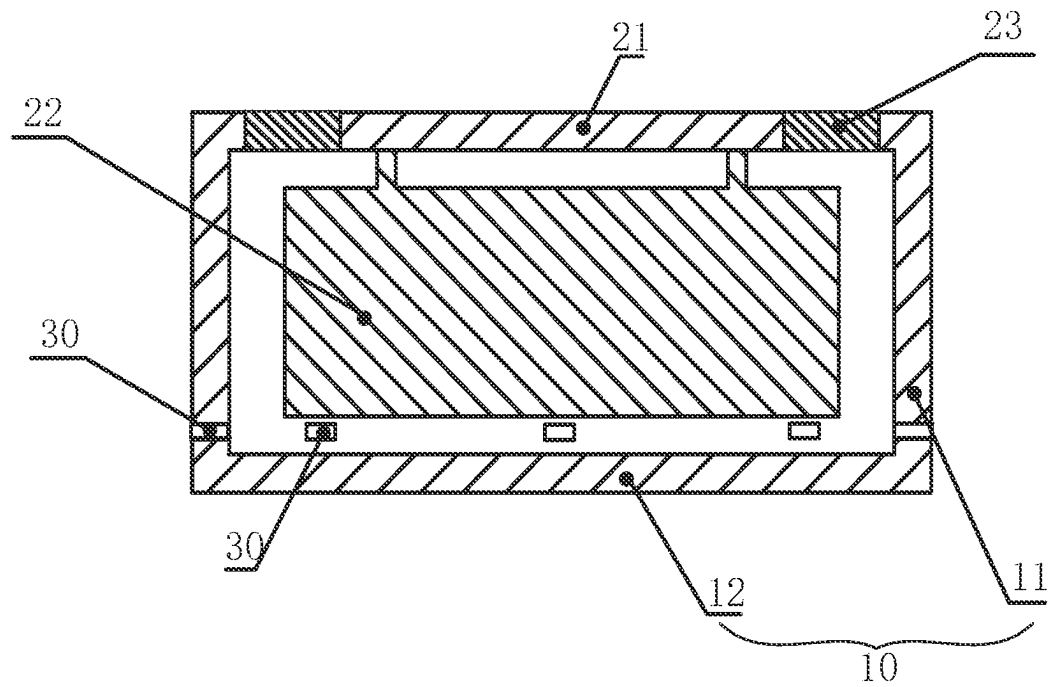


FIG. 7B

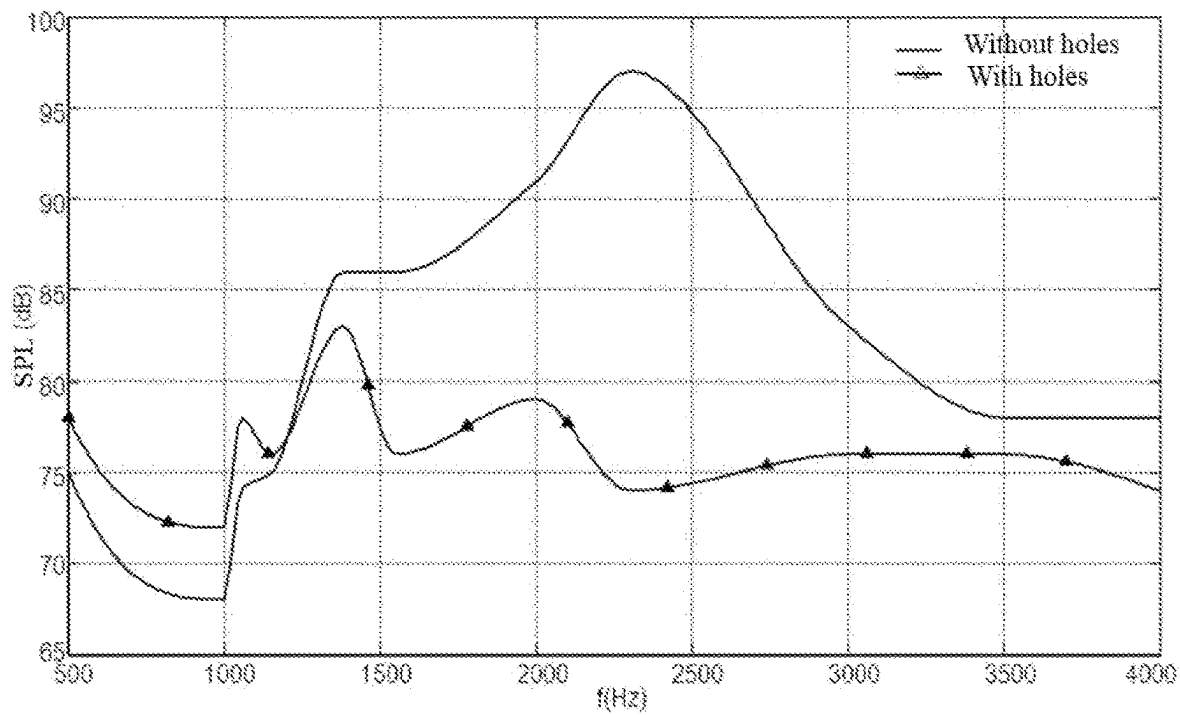


FIG. 7C

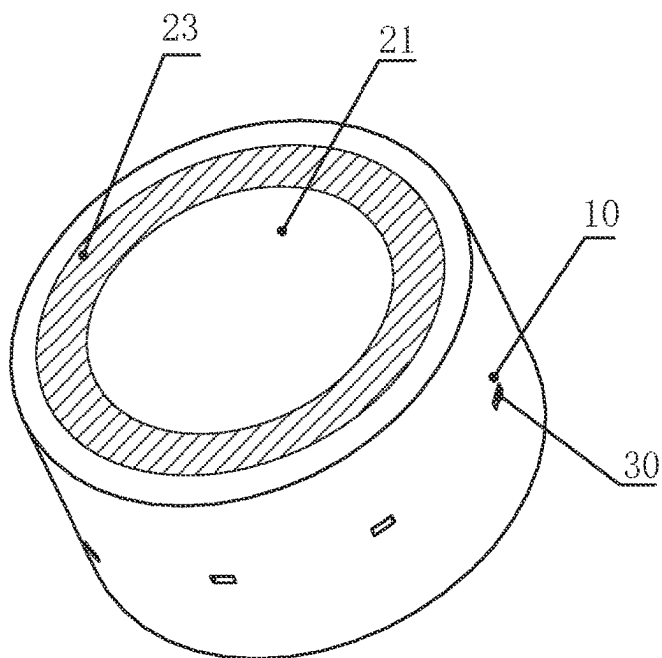


FIG. 8A

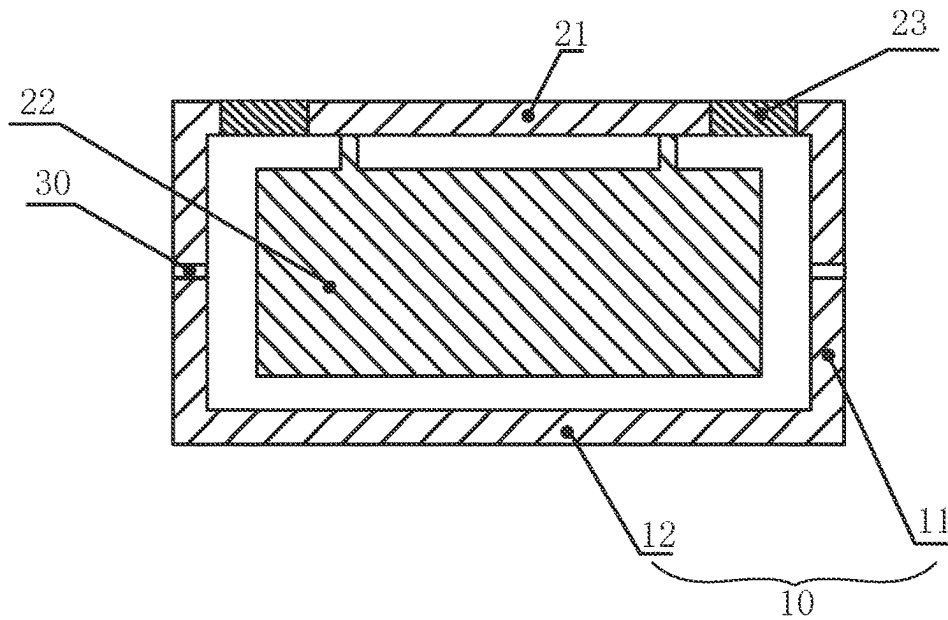


FIG. 8B

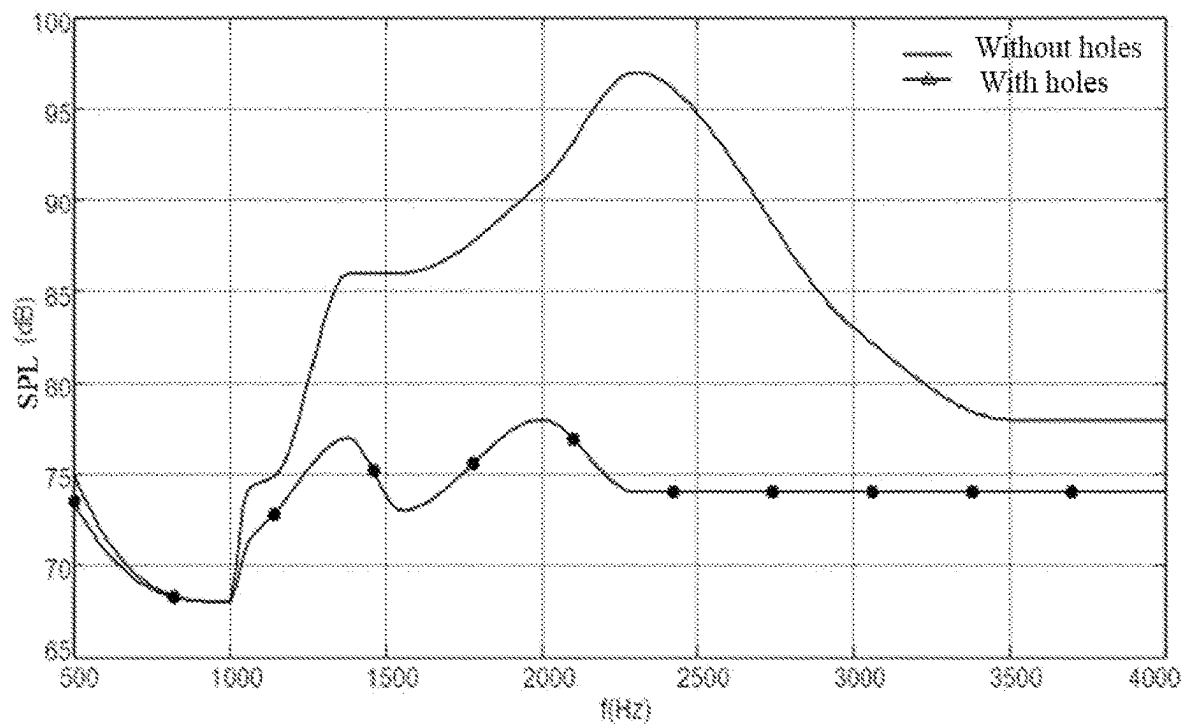


FIG. 8C

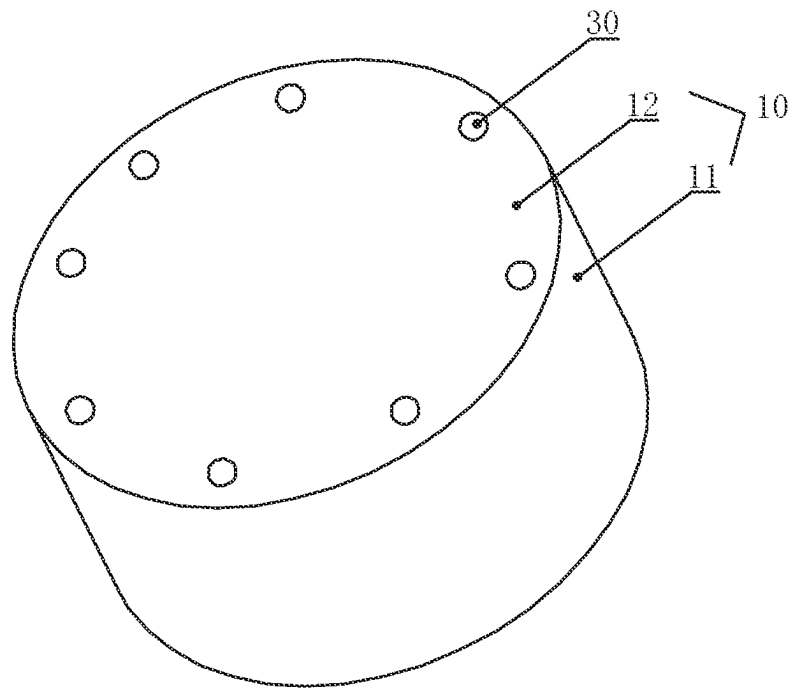


FIG. 9A

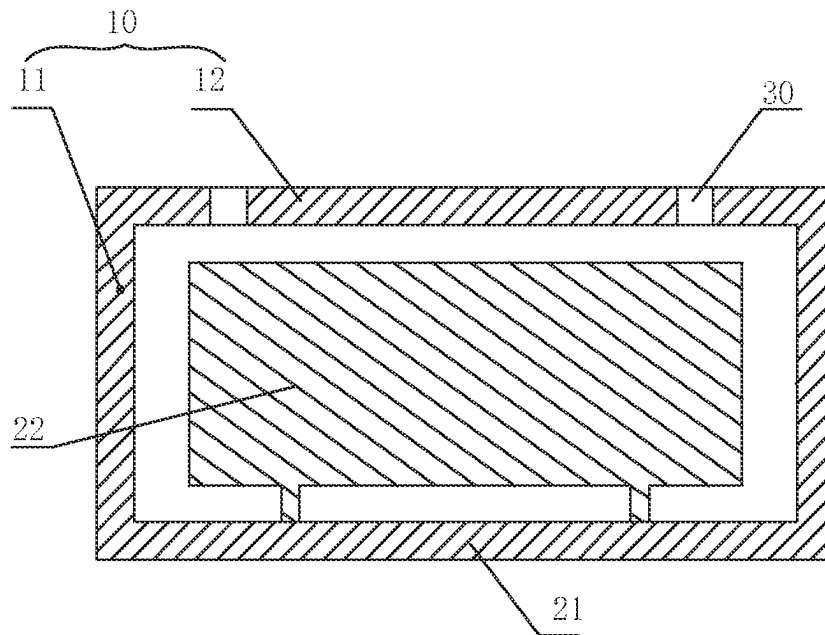


FIG. 9B

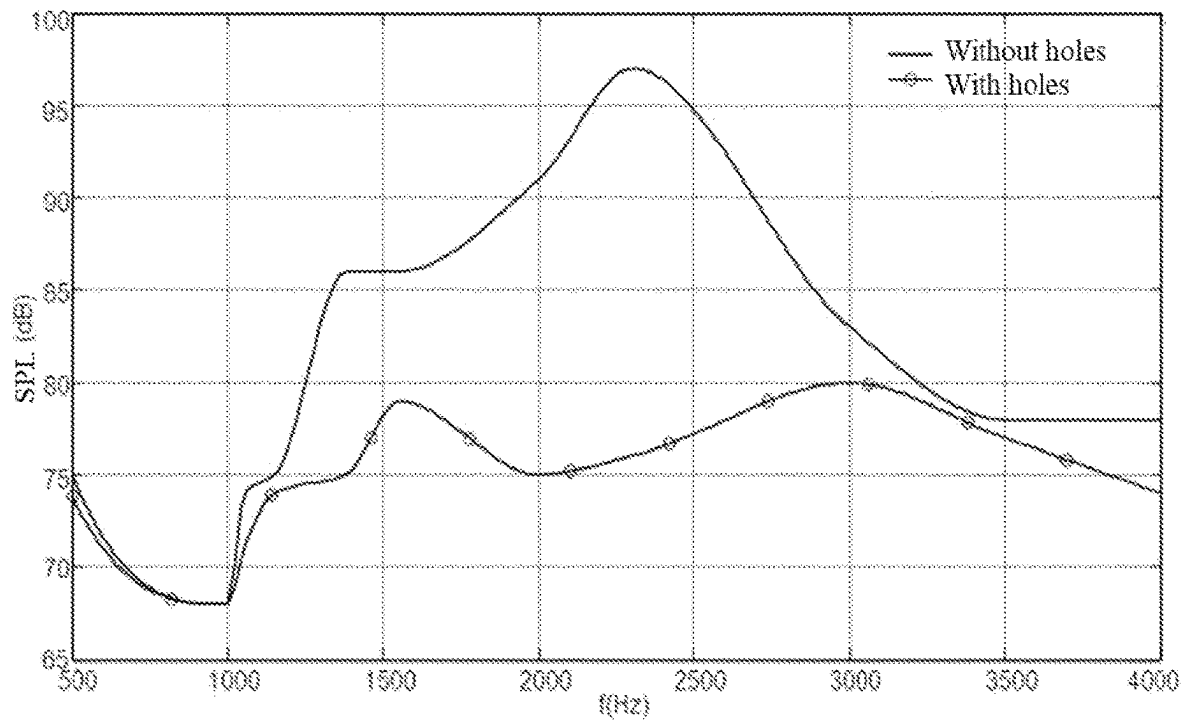


FIG. 9C

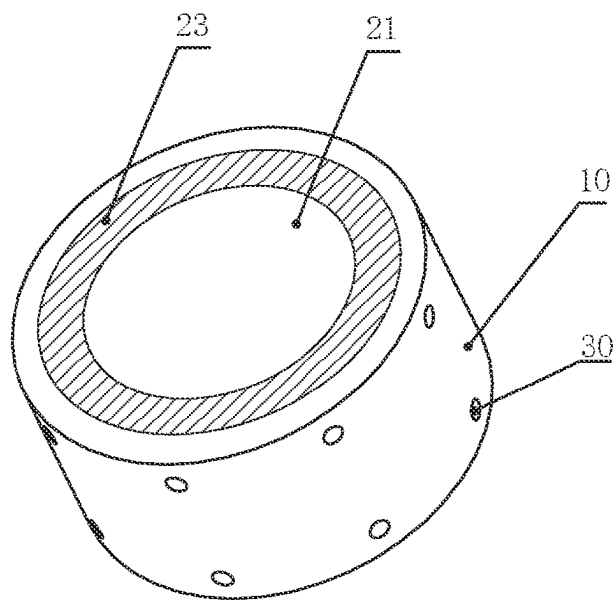


FIG. 10A

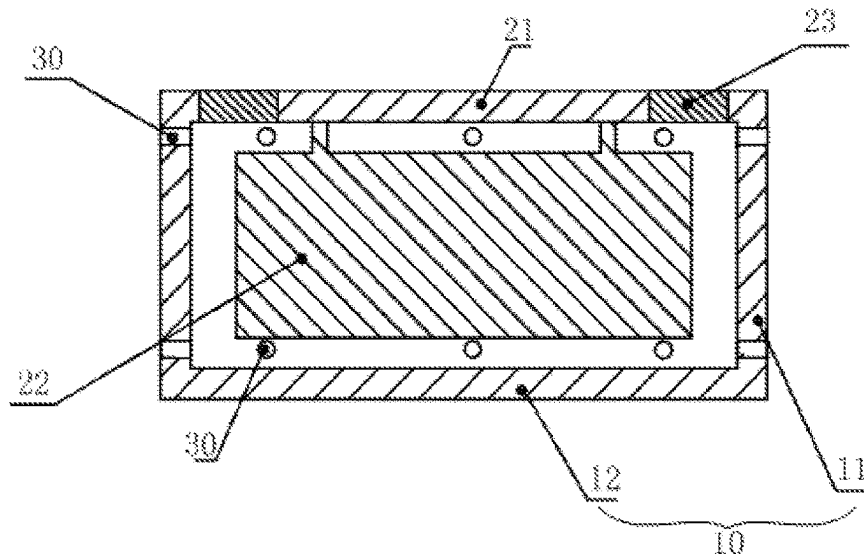


FIG. 10B

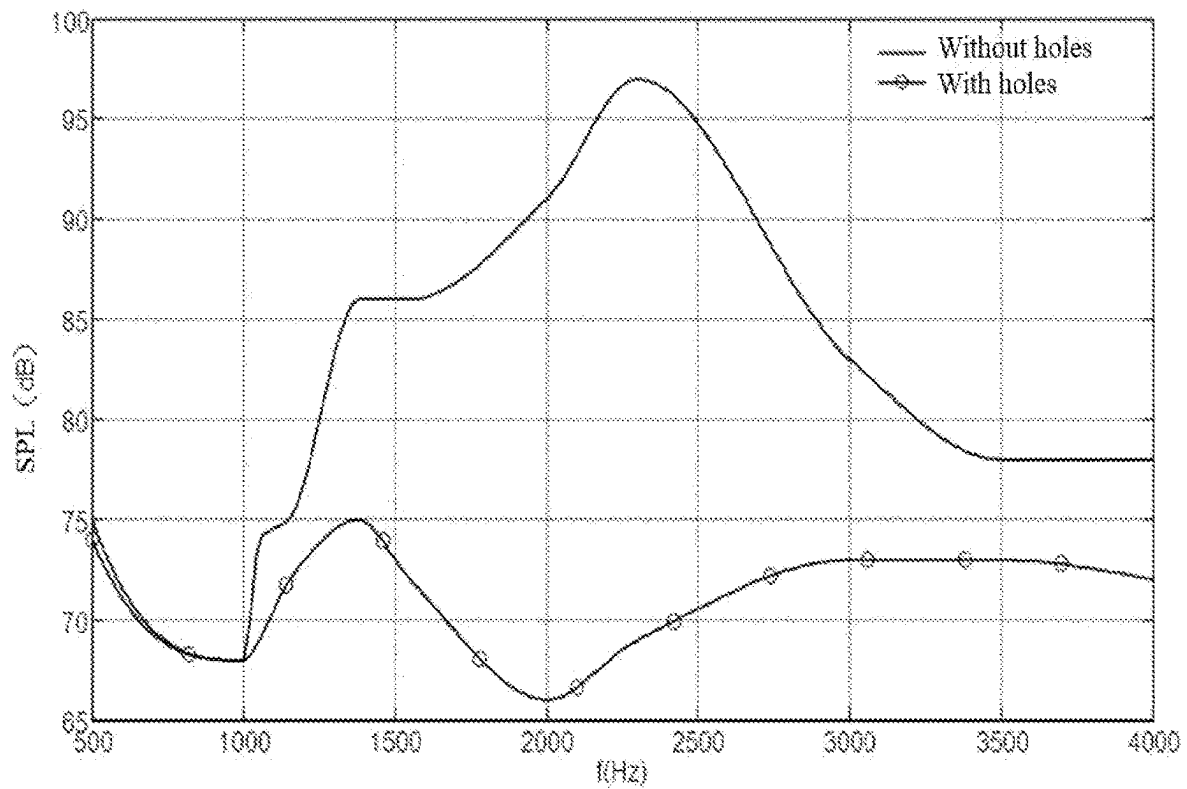


FIG. 10C

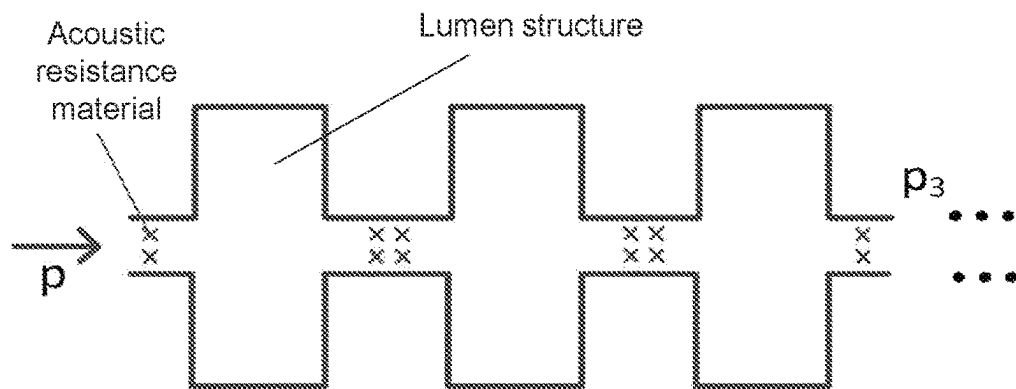


FIG. 10D

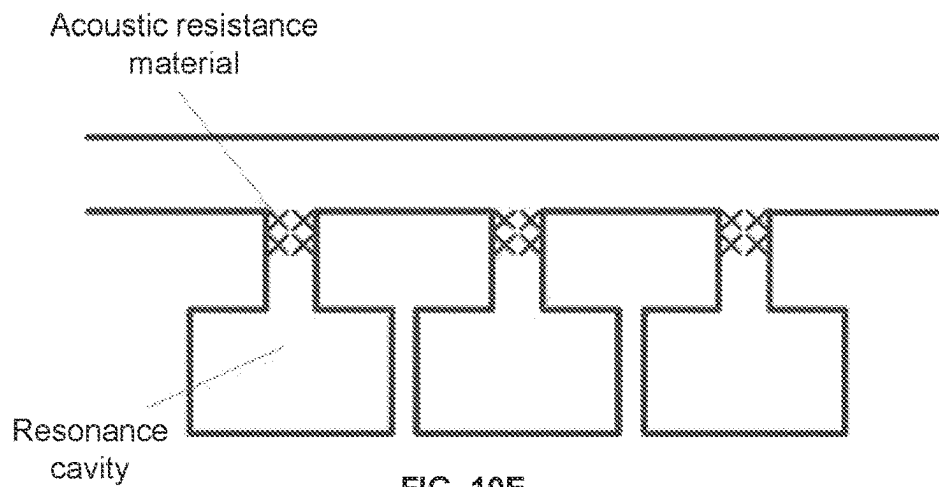


FIG. 10E

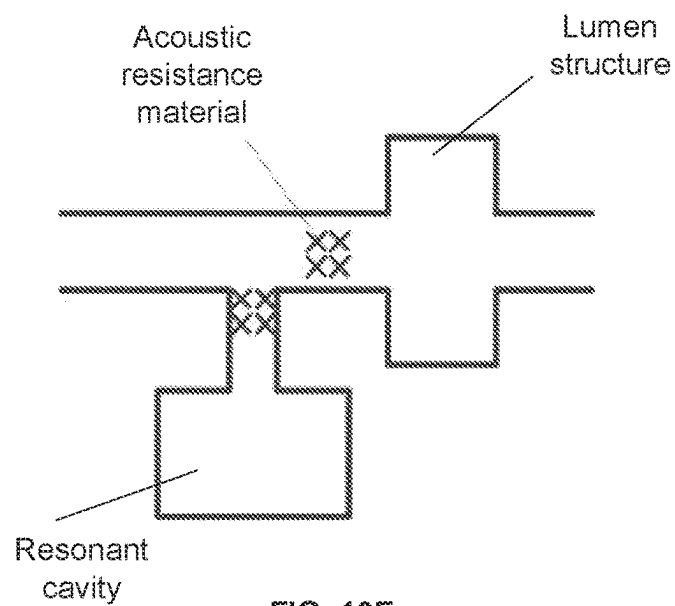


FIG. 10F

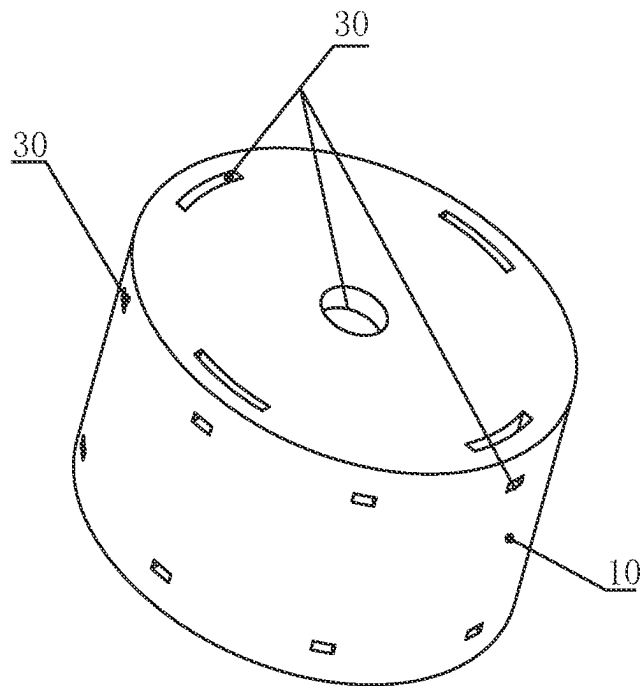


FIG. 11A

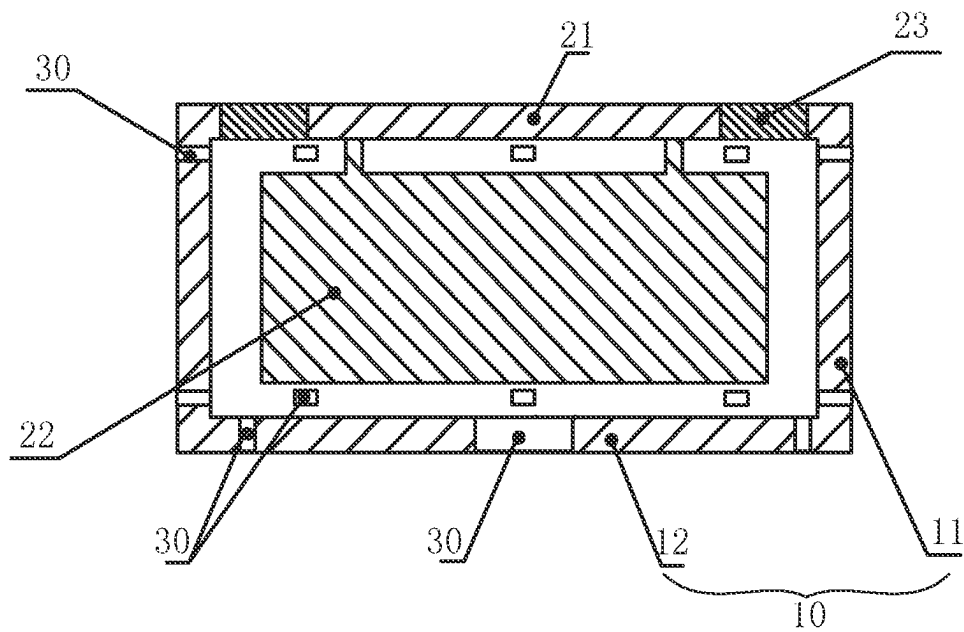


FIG. 11B

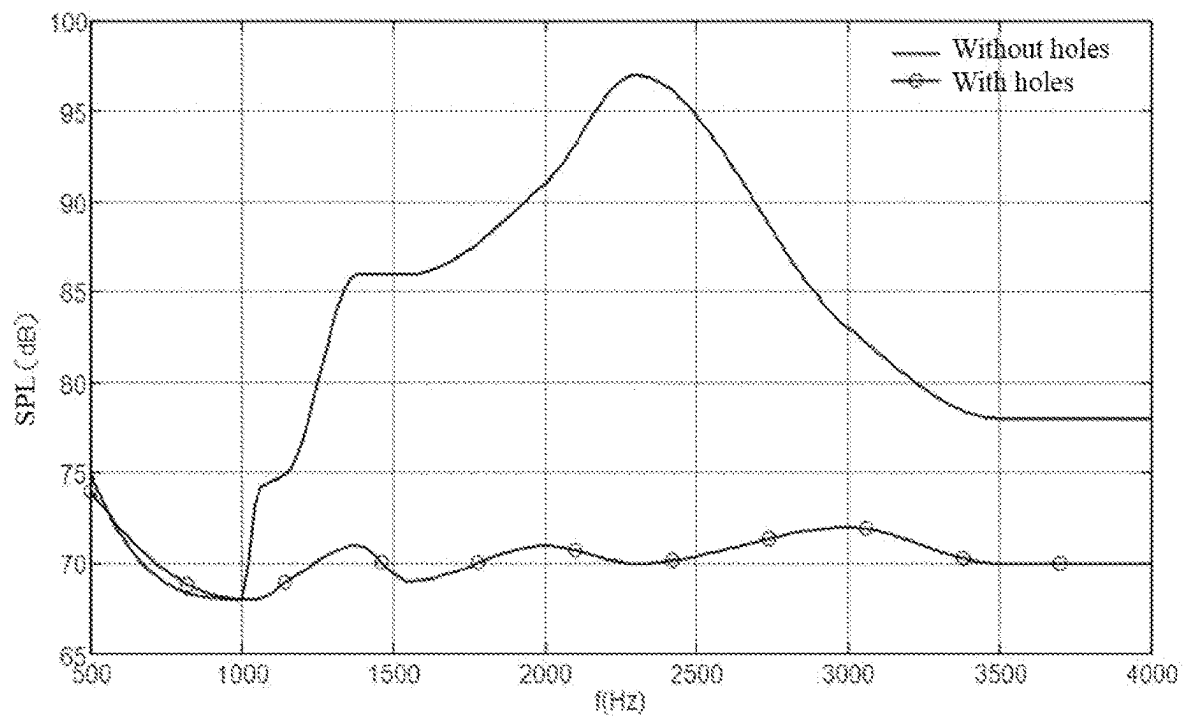


FIG. 11C

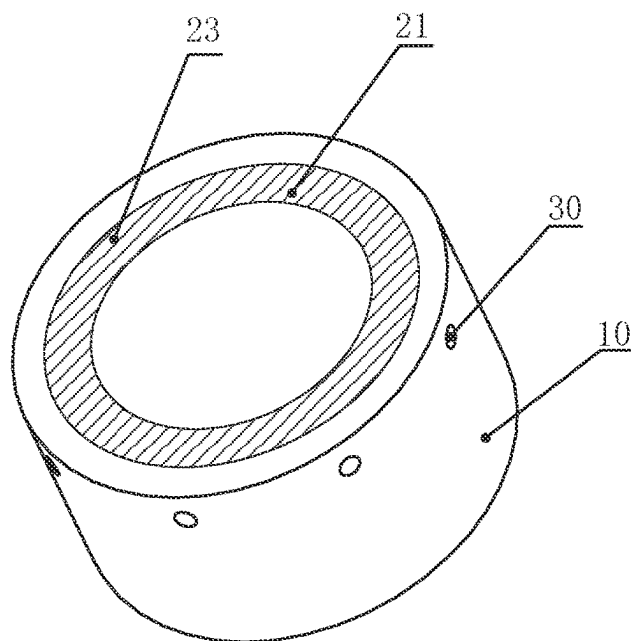
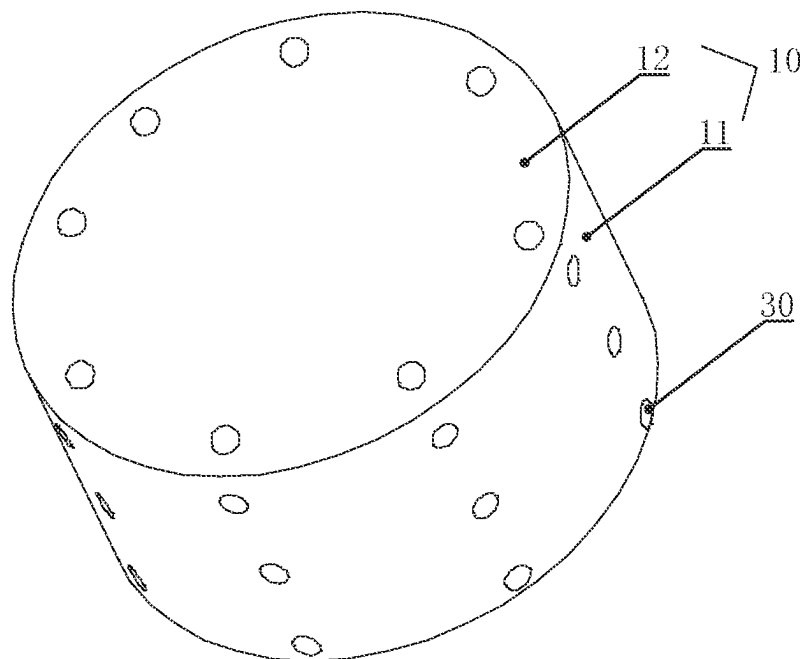
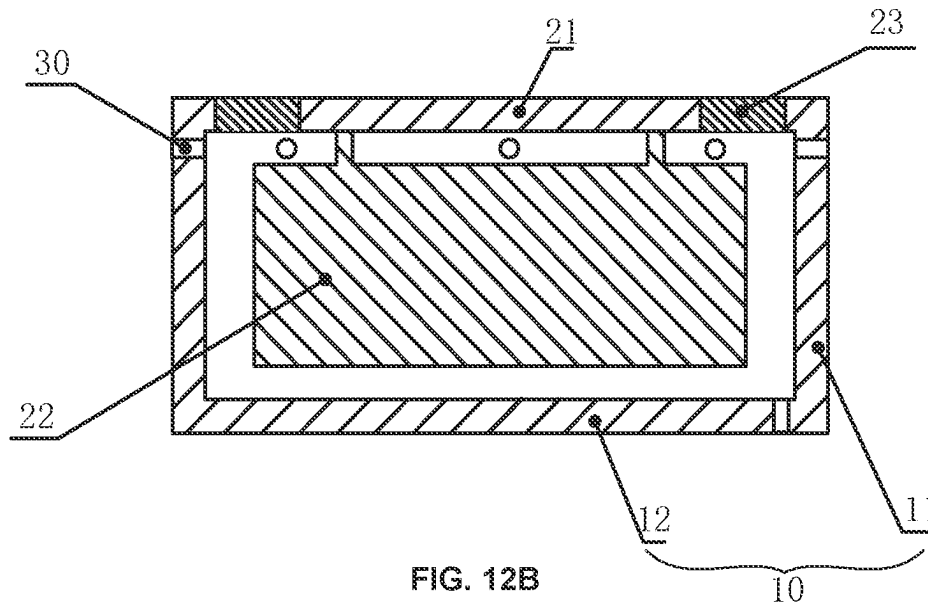


FIG. 12A



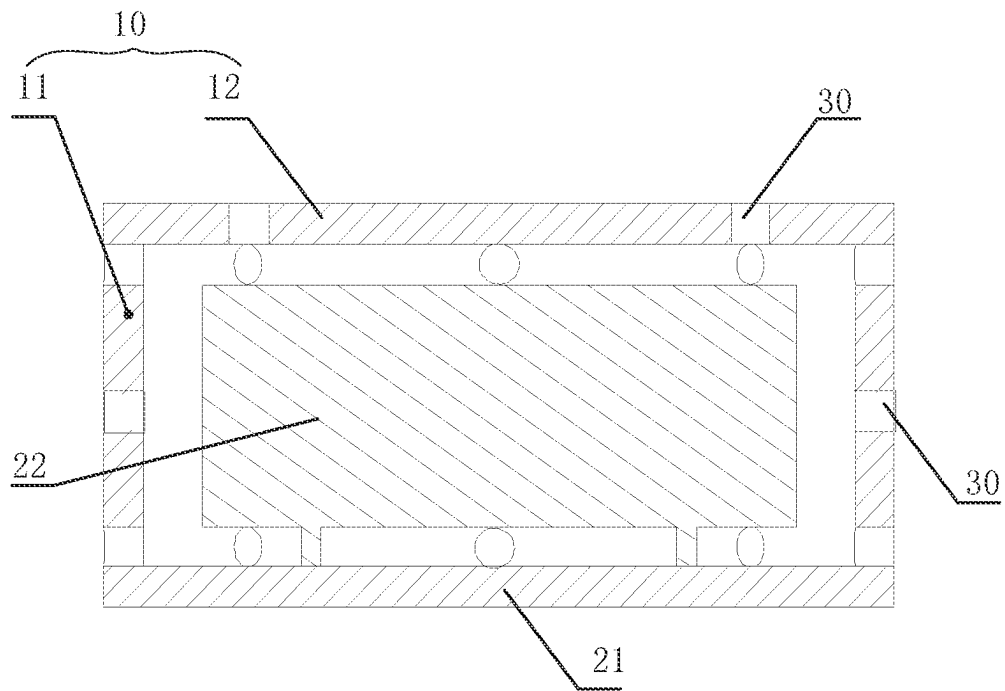


FIG. 13B

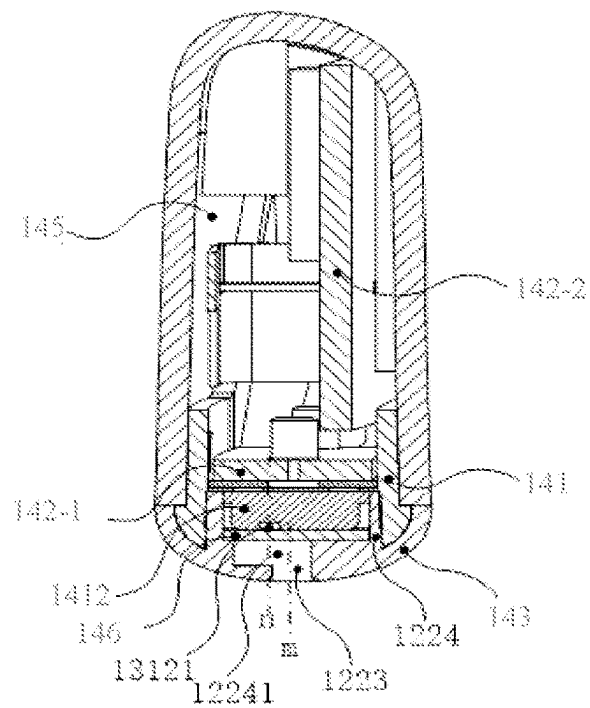


FIG. 14

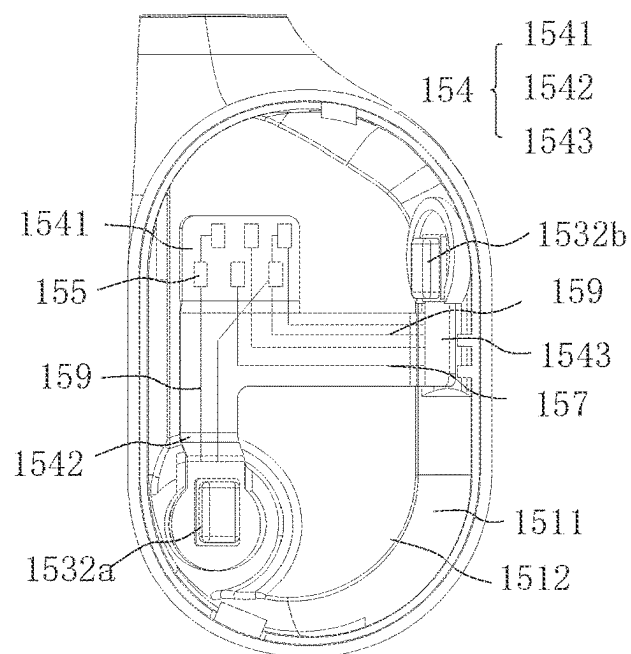


FIG. 15

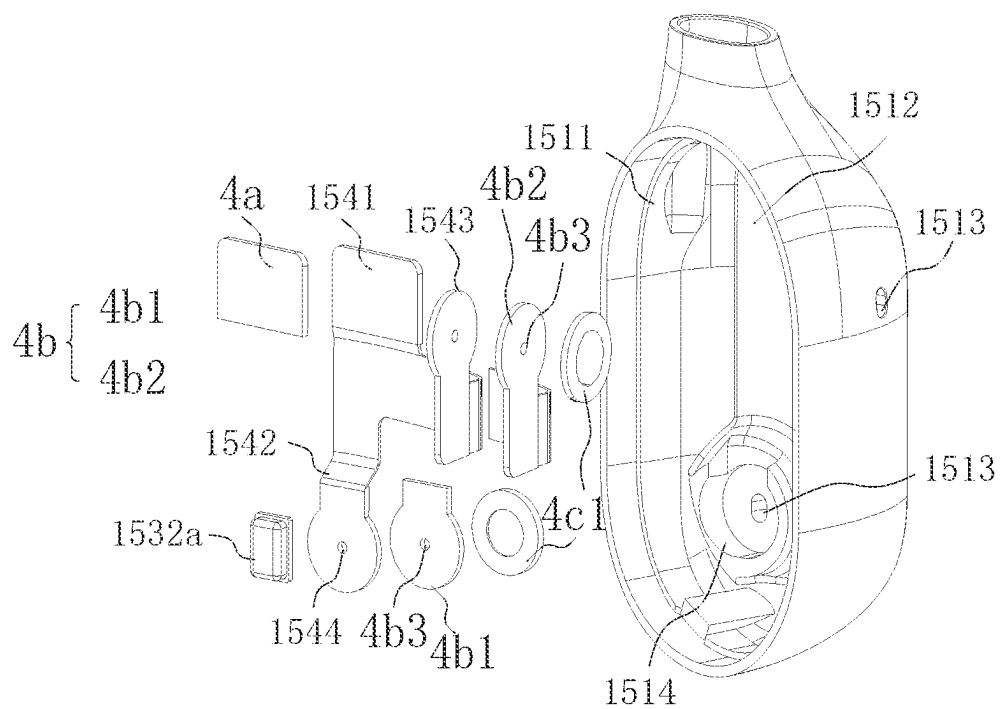


FIG. 16

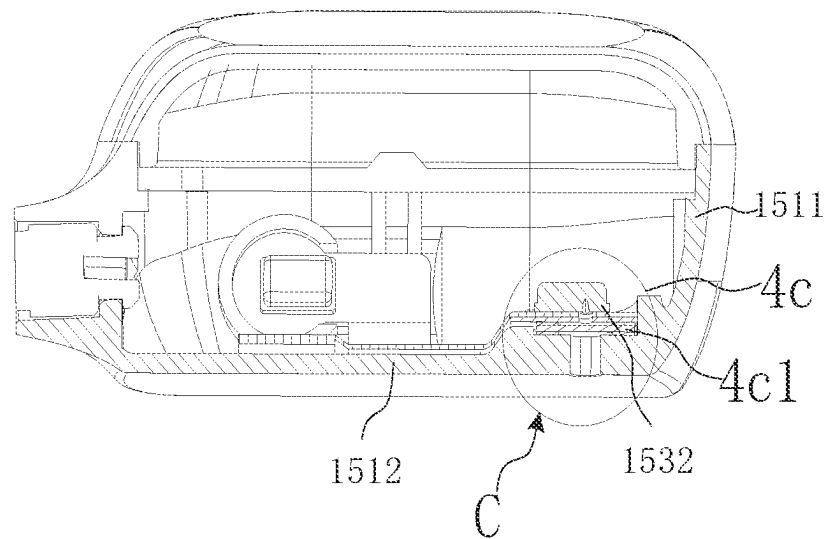


FIG. 17

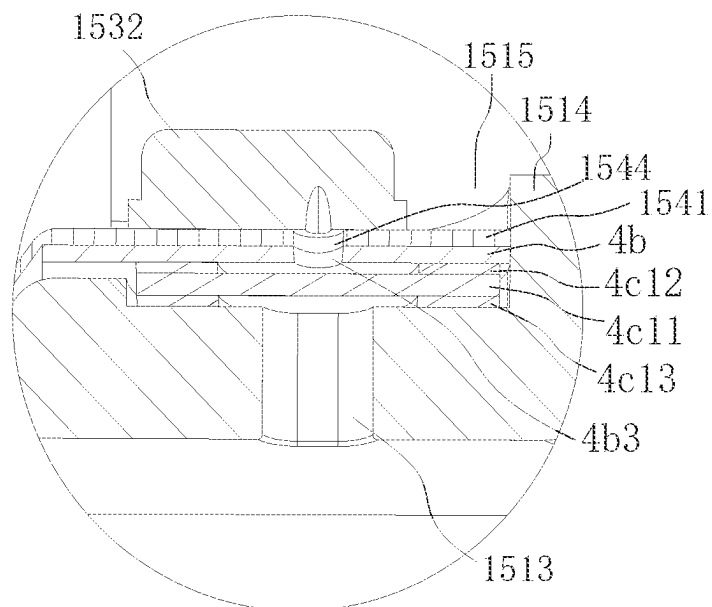


FIG. 18

1

SYSTEMS AND METHODS FOR SUPPRESSING SOUND LEAKAGE

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 17/171,987 filed on Feb. 9, 2021, which is a continuation-in-part of U.S. patent application Ser. No. 17/074,762 (Now U.S. Pat. No. 11,197,106) filed on Oct. 20, 2020, which is a continuation-in-part of U.S. patent application Ser. No. 16/813,915 (now U.S. Pat. No. 10,848,878) filed on Mar. 10, 2020, which is a continuation of U.S. patent application Ser. No. 16/419,049 (now U.S. Pat. No. 10,616,696) filed on May 22, 2019, which is a continuation of U.S. patent application Ser. No. 16/180,020 (now U.S. Pat. No. 10,334,372) filed on Nov. 5, 2018, which is a continuation of U.S. patent application Ser. No. 15/650,909 (now U.S. Pat. No. 10,149,071) filed on Jul. 16, 2017, which is a continuation of U.S. patent application Ser. No. 15/109,831 (now U.S. Pat. No. 9,729,978) filed on Jul. 6, 2016, which is a U.S. National Stage entry under 35 U.S.C. § 371 of International Application No. PCT/CN2014/094065, filed on Dec. 17, 2014, designating the United States of America, which claims priority to Chinese Patent Application No. 201410005804.0, filed on Jan. 6, 2014; U.S. patent application Ser. No. 17/171,987 is a continuation-in-part of U.S. patent application Ser. No. 16/950,876 filed on Nov. 17, 2020, which is a continuation of International Application No. PCT/CN2019/102394, filed on Aug. 24, 2019, which claims priority of Chinese Patent Application No. 201810975515.1 filed on Aug. 24, 2018. Each of the above-referenced applications is hereby incorporated by reference.

FIELD OF THE INVENTION

This application relates to a bone conduction device, and more specifically, relates to methods and systems for reducing sound leakage by a bone conduction device.

BACKGROUND

A bone conduction speaker, which may be also called a vibration speaker, may push human tissues and bones to stimulate the auditory nerve in cochlea and enable people to hear sound. The bone conduction speaker is also called a bone conduction headphone.

An exemplary structure of a bone conduction speaker based on the principle of the bone conduction speaker is shown in FIGS. 1A and 1B. The bone conduction speaker may include an open housing 110, a vibration board 121, a transducer 122, and a linking component 123. The transducer 122 may transduce electrical signals to mechanical vibrations. The vibration board 121 may be connected to the transducer 122 and vibrate synchronically with the transducer 122. The vibration board 121 may stretch out from the opening of the housing 110 and contact with human skin to pass vibrations to auditory nerves through human tissues and bones, which in turn enables people to hear sound. The linking component 123 may reside between the transducer 122 and the housing 110, configured to fix the vibrating transducer 122 inside the housing 110. To minimize its effect on the vibrations generated by the transducer 122, the linking component 123 may be made of an elastic material.

However, the mechanical vibrations generated by the transducer 122 may not only cause the vibration board 121 to vibrate, but may also cause the housing 110 to vibrate

2

through the linking component 123. Accordingly, the mechanical vibrations generated by the bone conduction speaker may push human tissues through the bone board 121, and at the same time a portion of the vibrating board 121 and the housing 110 that are not in contact with human issues may nevertheless push air. Air sound may thus be generated by the air pushed by the portion of the vibrating board 121 and the housing 110. The air sound may be called “sound leakage.” In some cases, sound leakage is harmless. However, sound leakage should be avoided as much as possible if people intend to protect privacy when using the bone conduction speaker or try not to disturb others when listening to music.

Attempting to solve the problem of sound leakage, Korean patent KR10-2009-0082999 discloses a bone conduction speaker of a dual magnetic structure and double-frame. As shown in FIG. 2, the speaker disclosed in the patent includes: a first frame 210 with an open upper portion and a second frame 220 that surrounds the outside of the first frame 210. The second frame 220 is separately placed from the outside of the first frame 210. The first frame 210 includes a movable coil 230 with electric signals, an inner magnetic component 240, an outer magnetic component 250, a magnet field formed between the inner magnetic component 240, and the outer magnetic component 250. The inner magnetic component 240 and the out magnetic component 250 may vibrate by the attraction and repulsion force of the coil 230 placed in the magnet field. A vibration board 260 connected to the moving coil 230 may receive the vibration of the moving coil 230. A vibration unit 270 connected to the vibration board 260 may pass the vibration to a user by contacting with the skin. As described in the patent, the second frame 220 surrounds the first frame 210, in order to use the second frame 220 to prevent the vibration of the first frame 210 from dissipating the vibration to outsides, and thus may reduce sound leakage to some extent.

However, in this design, since the second frame 220 is fixed to the first frame 210, vibrations of the second frame 220 are inevitable. As a result, sealing by the second frame 220 is unsatisfactory. Furthermore, the second frame 220 increases the whole volume and weight of the speaker, which in turn increases the cost, complicates the assembly process, and reduces the speaker’s reliability and consistency.

SUMMARY

The embodiments of the present application disclose methods and system of reducing sound leakage of a bone conduction speaker.

In one aspect, the embodiments of the present application disclose a method of reducing sound leakage of a bone conduction speaker, including:

providing a bone conduction speaker including a vibration board fitting human skin and passing vibrations, a transducer, and a housing, wherein at least one sound guiding hole is located in at least one portion of the housing;

the transducer drives the vibration board to vibrate; the housing vibrates, along with the vibrations of the transducer, and pushes air, forming a leaked sound wave transmitted in the air;

the air inside the housing is pushed out of the housing through the at least one sound guiding hole, interferes with the leaked sound wave, and reduces an amplitude of the leaked sound wave.

3

In some embodiments, one or more sound guiding holes may locate in an upper portion, a central portion, and/or a lower portion of a sidewall and/or the bottom of the housing.

In some embodiments, a damping layer may be applied in the at least one sound guiding hole in order to adjust the phase and amplitude of the guided sound wave through the at least one sound guiding hole.

In some embodiments, sound guiding holes may be configured to generate guided sound waves having a same phase that reduce the leaked sound wave having a same wavelength; sound guiding holes may be configured to generate guided sound waves having different phases that reduce the leaked sound waves having different wavelengths.

In some embodiments, different portions of a same sound guiding hole may be configured to generate guided sound waves having a same phase that reduce the leaked sound wave having same wavelength. In some embodiments, different portions of a same sound guiding hole may be configured to generate guided sound waves having different phases that reduce leaked sound waves having different wavelengths.

In another aspect, the embodiments of the present application disclose a bone conduction speaker, including a housing, a vibration board and a transducer, wherein: the transducer is configured to generate vibrations and is located inside the housing; the vibration board is configured to be in contact with skin and pass vibrations.

At least one sound guiding hole may locate in at least one portion on the housing, and preferably, the at least one sound guiding hole may be configured to guide a sound wave inside the housing, resulted from vibrations of the air inside the housing, to the outside of the housing, the guided sound wave interfering with the leaked sound wave and reducing the amplitude thereof.

In some embodiments, the at least one sound guiding hole may locate in the sidewall and/or bottom of the housing.

In some embodiments, preferably, the at least one sound guiding hole may locate in the upper portion and/or lower portion of the sidewall of the housing.

In some embodiments, preferably, the sidewall of the housing is cylindrical and there are at least two sound guiding holes located in the sidewall of the housing, which are arranged evenly or unevenly in one or more circles. Alternatively, the housing may have a different shape.

In some embodiments, preferably, the sound guiding holes have different heights along the axial direction of the cylindrical sidewall.

In some embodiments, preferably, there are at least two sound guiding holes located in the bottom of the housing. In some embodiments, the sound guiding holes are distributed evenly or unevenly in one or more circles around the center of the bottom. Alternatively or additionally, one sound guiding hole is located at the center of the bottom of the housing.

In some embodiments, preferably, the sound guiding hole is a perforative hole. In some embodiments, there may be a damping layer at the opening of the sound guiding hole.

In some embodiments, preferably, the guided sound waves through different sound guiding holes and/or different portions of a same sound guiding hole have different phases or a same phase.

In some embodiments, preferably, the damping layer is a tuning paper, a tuning cotton, a nonwoven fabric, a silk, a cotton, a sponge, or a rubber.

In some embodiments, preferably, the shape of a sound guiding hole is circle, ellipse, quadrangle, rectangle, or

4

linear. In some embodiments, the sound guiding holes may have a same shape or different shapes.

In some embodiments, preferably, the transducer includes a magnetic component and a voice coil. Alternatively, the transducer includes piezoelectric ceramic.

The design disclosed in this application utilizes the principles of sound interference, by placing sound guiding holes in the housing, to guide sound wave(s) inside the housing to the outside of the housing, the guided sound wave(s) interfering with the leaked sound wave, which is formed when the housing's vibrations push the air outside the housing. The guided sound wave(s) reduces the amplitude of the leaked sound wave and thus reduces the sound leakage. The design not only reduces sound leakage, but is also easy to implement, doesn't increase the volume or weight of the bone conduction speaker, and barely increase the cost of the product.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A and 1B are schematic structures illustrating a bone conduction speaker of prior art;

FIG. 2 is a schematic structure illustrating another bone conduction speaker of prior art;

FIG. 3 illustrates the principle of sound interference according to some embodiments of the present disclosure;

FIGS. 4A and 4B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIG. 4C is a schematic structure of the bone conduction speaker according to some embodiments of the present disclosure;

FIG. 4D is a diagram illustrating reduced sound leakage of the bone conduction speaker according to some embodiments of the present disclosure;

FIG. 4E is a schematic diagram illustrating exemplary two-point sound sources according to some embodiments of the present disclosure;

FIG. 5 is a diagram illustrating the equal-loudness contour curves according to some embodiments of the present disclosure;

FIG. 6 is a flow chart of an exemplary method of reducing sound leakage of a bone conduction speaker according to some embodiments of the present disclosure;

FIGS. 7A and 7B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIG. 7C is a diagram illustrating reduced sound leakage of a bone conduction speaker according to some embodiments of the present disclosure;

FIGS. 8A and 8B are schematic structure of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIG. 8C is a diagram illustrating reduced sound leakage of a bone conduction speaker according to some embodiments of the present disclosure;

FIGS. 9A and 9B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIG. 9C is a diagram illustrating reduced sound leakage of a bone conduction speaker according to some embodiments of the present disclosure;

FIGS. 10A and 10B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

5

FIG. 10C is a diagram illustrating reduced sound leakage of a bone conduction speaker according to some embodiments of the present disclosure;

FIG. 10D is a schematic diagram illustrating an acoustic route according to some embodiments of the present disclosure;

FIG. 10E is a schematic diagram illustrating another acoustic route according to some embodiments of the present disclosure;

FIG. 10F is a schematic diagram illustrating a further acoustic route according to some embodiments of the present disclosure;

FIGS. 11A and 11B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIG. 11C is a diagram illustrating reduced sound leakage of a bone conduction speaker according to some embodiments of the present disclosure; and

FIGS. 12A and 12B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIGS. 13A and 13B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIG. 14 is a sectional view illustrating an electronic component according to some embodiments of the present disclosure;

FIG. 15 is a partial structural diagram illustrating a speaker according to some embodiments of the present disclosure;

FIG. 16 is an exploded view illustrating a partial structure of a speaker according to some embodiments of the present disclosure;

FIG. 17 is a sectional view illustrating a partial structure of a speaker according to some embodiments of the present disclosure; and

FIG. 18 is a partial enlarged view illustrating part C in FIG. 17 according to some embodiments of the present disclosure.

The meanings of the mark numbers in the figures are as followed:

110, open housing; 121, vibration board; 122, transducer; 123, linking component; 210, first frame; 220, second frame; 230, moving coil; 240, inner magnetic component; 250, outer magnetic component; 260; vibration board; 270, vibration unit; 10, housing; 11, sidewall; 12, bottom; 21, vibration board; 22, transducer; 23, linking component; 24, elastic component; 30, sound guiding hole.

DETAILED DESCRIPTION

Followings are some further detailed illustrations about this disclosure. The following examples are for illustrative purposes only and should not be interpreted as limitations of the claimed invention. There are a variety of alternative techniques and procedures available to those of ordinary skill in the art, which would similarly permit one to successfully perform the intended invention. In addition, the figures just show the structures relative to this disclosure, not the whole structure.

To explain the scheme of the embodiments of this disclosure, the design principles of this disclosure will be introduced here. FIG. 3 illustrates the principles of sound interference according to some embodiments of the present disclosure. Two or more sound waves may interfere in the space based on, for example, the frequency and/or amplitude

6

of the waves. Specifically, the amplitudes of the sound waves with the same frequency may be overlaid to generate a strengthened wave or a weakened wave. As shown in FIG. 3, sound source 1 and sound source 2 have the same frequency and locate in different locations in the space. The sound waves generated from these two sound sources may encounter in an arbitrary point A. If the phases of the sound wave 1 and sound wave 2 are the same at point A, the amplitudes of the two sound waves may be added, generating a strengthened sound wave signal at point A; on the other hand, if the phases of the two sound waves are opposite at point A, their amplitudes may be offset, generating a weakened sound wave signal at point A.

This disclosure applies above-noted the principles of sound wave interference to a bone conduction speaker and disclose a bone conduction speaker that can reduce sound leakage.

Embodiment One

FIGS. 4A and 4B are schematic structures of an exemplary bone conduction speaker. The bone conduction speaker may include a housing 10, a vibration board 21, and a transducer 22. The transducer 22 may be inside the housing 10 and configured to generate vibrations. The housing 10 may have one or more sound guiding holes 30. The sound guiding hole(s) 30 may be configured to guide sound waves inside the housing 10 to the outside of the housing 10. In some embodiments, the guided sound waves may form interference with leaked sound waves generated by the vibrations of the housing 10, so as to reducing the amplitude of the leaked sound. The transducer 22 may be configured to convert an electrical signal to mechanical vibrations. For example, an audio electrical signal may be transmitted into a voice coil that is placed in a magnet, and the electromagnetic interaction may cause the voice coil to vibrate based on the audio electrical signal. As another example, the transducer 22 may include piezoelectric ceramics, shape changes of which may cause vibrations in accordance with electrical signals received.

Furthermore, the vibration board 21 may be connected to the transducer 22 and configured to vibrate along with the transducer 22. The vibration board 21 may stretch out from the opening of the housing 10, and touch the skin of the user and pass vibrations to auditory nerves through human tissues and bones, which in turn enables the user to hear sound. The linking component 23 may reside between the transducer 22 and the housing 10, configured to fix the vibrating transducer 122 inside the housing. The linking component 23 may include one or more separate components, or may be integrated with the transducer 22 or the housing 10. In some embodiments, the linking component 23 is made of an elastic material.

The transducer 22 may drive the vibration board 21 to vibrate. The transducer 22, which resides inside the housing 10, may vibrate. The vibrations of the transducer 22 may drives the air inside the housing 10 to vibrate, producing a sound wave inside the housing 10, which can be referred to as "sound wave inside the housing." Since the vibration board 21 and the transducer 22 are fixed to the housing 10 via the linking component 23, the vibrations may pass to the housing 10, causing the housing 10 to vibrate synchronously. The vibrations of the housing 10 may generate a leaked sound wave, which spreads outwards as sound leakage.

The sound wave inside the housing and the leaked sound wave are like the two sound sources in FIG. 3. In some

7

embodiments, the sidewall **11** of the housing **10** may have one or more sound guiding holes **30** configured to guide the sound wave inside the housing **10** to the outside. The guided sound wave through the sound guiding hole(s) **30** may interfere with the leaked sound wave generated by the vibrations of the housing **10**, and the amplitude of the leaked sound wave may be reduced due to the interference, which may result in a reduced sound leakage. Therefore, the design of this embodiment can solve the sound leakage problem to some extent by making an improvement of setting a sound guiding hole on the housing, and not increasing the volume and weight of the bone conduction speaker.

In some embodiments, one sound guiding hole **30** is set on the upper portion of the sidewall **11**. As used herein, the upper portion of the sidewall **11** refers to the portion of the sidewall **11** starting from the top of the sidewall (contacting with the vibration board **21**) to about the $\frac{1}{3}$ height of the sidewall.

FIG. **4C** is a schematic structure of the bone conduction speaker illustrated in FIGS. **4A-4B**. The structure of the bone conduction speaker is further illustrated with mechanics elements illustrated in FIG. **4C**. As shown in FIG. **4C**, the linking component **23** between the sidewall **11** of the housing **10** and the vibration board **21** may be represented by an elastic element **23** and a damping element in the parallel connection. The linking relationship between the vibration board **21** and the transducer **22** may be represented by an elastic element **24**.

Outside the housing **10**, the sound leakage reduction is proportional to

$$(\iint_{S_{hole}} P_{ds} - \iint_{S_{housing}} P_{ads}), \quad (1)$$

wherein S_{hole} is the area of the opening of the sound guiding hole **30**, $S_{housing}$ is the area of the housing **10** (e.g., the sidewall **11** and the bottom **12**) that is not in contact with human face.

The pressure inside the housing may be expressed as

$$P = P_a + P_b + P_c + P_e, \quad (2)$$

wherein P_a , P_b , P_c and P_e are the sound pressures of an arbitrary point inside the housing **10** generated by side a, side b, side c and side e (as illustrated in FIG. **4C**), respectively. As used herein, side a refers to the upper surface of the transducer **22** that is close to the vibration board **21**, side b refers to the lower surface of the vibration board **21** that is close to the transducer **22**, side c refers to the inner upper surface of the bottom **12** that is close to the transducer **22**, and side e refers to the lower surface of the transducer **22** that is close to the bottom **12**.

The center of the side b, O point, is set as the origin of the space coordinates, and the side b can be set as the $z=0$ plane, so P_a , P_b , P_c and P_e may be expressed as follows:

$$P_a(x, y, z) = -j\omega\rho_0 \iint_{S_a} W_a(x'_a, y'_a) \cdot \frac{e^{jkR(x'_a, y'_a)}}{4\pi R(x'_a, y'_a)} dx'_a dy'_a - P_{aR}, \quad (3)$$

$$P_b(x, y, z) = -j\omega\rho_0 \iint_{S_b} W_b(x', y') \cdot \frac{e^{jkR(x', y')}}{4\pi R(x', y')} dx' dy' - P_{bR}, \quad (4)$$

$$P_c(x, y, z) = -j\omega\rho_0 \iint_{S_c} W_c(x'_c, y'_c) \cdot \frac{e^{jkR(x'_c, y'_c)}}{4\pi R(x'_c, y'_c)} dx'_c dy'_c - P_{cR}, \quad (5)$$

$$P_e(x, y, z) = -j\omega\rho_0 \iint_{S_e} W_e(x'_e, y'_e) \cdot \frac{e^{jkR(x'_e, y'_e)}}{4\pi R(x'_e, y'_e)} dx'_e dy'_e - P_{eR}, \quad (6)$$

8

wherein $R(x', y') = \sqrt{(x-x')^2 + (y-y')^2 + z^2}$ is the distance between an observation point (x, y, z) and a point on side b $(x', y', 0)$; S_a , S_b , S_c and S_e are the areas of side a, side b, side c and side e, respectively;

$R(x'_a, y'_a) = \sqrt{(x-x'_a)^2 + (y-y'_a)^2 + (z-z'_a)^2}$ is the distance between the observation point (x, y, z) and a point on side a (x'_a, y'_a, z'_a) ;

$R(x'_c, y'_c) = \sqrt{(x-x'_c)^2 + (y-y'_c)^2 + (z-z'_c)^2}$ is the distance between the observation point (x, y, z) and a point on side c (x'_c, y'_c, z'_c) ;

$R(x'_e, y'_e) = \sqrt{(x-x'_e)^2 + (y-y'_e)^2 + (z-z'_e)^2}$ is the distance between the observation point (x, y, z) and a point on side e (x'_e, y'_e, z'_e) ;

$k = \omega/u$ (u is the velocity of sound) is wave number, ρ_0 is an air density, ω is an angular frequency of vibration.

P_{aR} , P_{bR} , P_{cR} and P_{eR} are acoustic resistances of air, which respectively are:

$$P_{aR} = A \cdot \frac{z_a \cdot r + j\omega \cdot z_a \cdot r'}{\varphi} + \delta, \quad (7)$$

$$P_{bR} = A \cdot \frac{z_b \cdot r + j\omega \cdot z_b \cdot r'}{\varphi} + \delta, \quad (8)$$

$$P_{cR} = A \cdot \frac{z_c \cdot r + j\omega \cdot z_c \cdot r'}{\varphi} + \delta, \quad (9)$$

$$P_{eR} = A \cdot \frac{z_e \cdot r + j\omega \cdot z_e \cdot r'}{\varphi} + \delta, \quad (10)$$

wherein r is the acoustic resistance per unit length, r' is the sound quality per unit length, z_a is the distance between the observation point and side a, z_b is the distance between the observation point and side b, z_c is the distance between the observation point and side c, z_e is the distance between the observation point and side e.

$W_a(x, y)$, $W_b(x, y)$, $W_c(x, y)$, $W_e(x, y)$ and $W_d(x, y)$ are the sound source power per unit area of side a, side b, side c, side e and side d, respectively, which can be derived from following formulas (11):

$$F_a = F_d = F - k_1 \cos \omega t - \iint_{S_d} W_d(x, y) dx dy - \iint_{S_e} W_e(x, y) dx dy - f$$

$$F_b = -F + k_1 \cos \omega t + \iint_{S_b} W_b(x, y) dx dy - \iint_{S_e} W_e(x, y) dx dy - L$$

$$F_c = F_d = F_b - k_2 \cos \omega t - \iint_{S_c} W_c(x, y) dx dy - f - \gamma$$

$$F_d = F_b - k_2 \cos \omega t - \iint_{S_d} W_d(x, y) dx dy \quad (11)$$

wherein F is the driving force generated by the transducer **22**, F_a , F_b , F_c , F_d and F_e are the driving forces of side a, side b, side c, side d and side e, respectively. As used herein, side d is the outside surface of the bottom **12**. S_d is the region of side d, f is the viscous resistance formed in the small gap of the sidewalls, and $f = \eta \Delta s (dv/dy)$.

L is the equivalent load on human face when the vibration board acts on the human face, y is the energy dissipated on elastic element **24**, k_1 and k_2 are the elastic coefficients of elastic element **23** and elastic element **24** respectively, η is the fluid viscosity coefficient, dv/dy is the velocity gradient of fluid, Δs is the cross-section area of a subject (board), A is the amplitude, φ is the region of the sound field, and δ is a high order minimum (which is generated by the incompletely symmetrical shape of the housing).

The sound pressure of an arbitrary point outside the housing, generated by the vibration of the housing **10** is expressed as:

$$P_d = -j\omega\rho_0 \iint W_d(x'_d, y'_d) \cdot \frac{e^{j\sqrt{(x-x'_d)^2+(y-y'_d)^2+(z-z'_d)^2}}}{4\pi R(x'_d, y'_d)} dx'_d dy'_d; \quad (12)$$

wherein $R(x'_d, y'_d) = \sqrt{(x-x'_d)^2+(y-y'_d)^2+(z-z'_d)^2}$ is the distance between the observation point (x, y, z) and a point on side d (x'_d, y'_d, z'_d) .

P_a, P_b, P_c and P_e are functions of the position, when we set a hole on an arbitrary position in the housing, if the area of the hole is S_{hole} , the sound pressure of the hole is $\iint_{S_{hole}} P_d ds$.

In the meanwhile, because the vibration board **21** fits human tissues tightly, the power it gives out is absorbed all by human tissues, so the only side that can push air outside the housing to vibrate is side d , thus forming sound leakage. As described elsewhere, the sound leakage is resulted from the vibrations of the housing **10**. For illustrative purposes, the sound pressure generated by the housing **10** may be expressed as $\iint_{S_{housing}} P_d ds$.

The leaked sound wave and the guided sound wave interference may result in a weakened sound wave, i.e., to make $\iint_{S_{hole}} P_d ds$ and $\iint_{S_{housing}} P_d ds$ have the same value but opposite directions, and the sound leakage may be reduced. In some embodiments, $\iint_{S_{hole}} P_d ds$ may be adjusted to reduce the sound leakage. Since $\iint_{S_{hole}} P_d ds$ corresponds to information of phases and amplitudes of one or more holes, which further relates to dimensions of the housing of the bone conduction speaker, the vibration frequency of the transducer, the position, shape, quantity and/or size of the sound guiding holes and whether there is damping inside the holes. Thus, the position, shape, and quantity of sound guiding holes, and/or damping materials may be adjusted to reduce sound leakage.

According to the formulas above, a person having ordinary skill in the art would understand that the effectiveness of reducing sound leakage is related to the dimensions of the housing of the bone conduction speaker, the vibration frequency of the transducer, the position, shape, quantity and size of the sound guiding hole(s) and whether there is damping inside the sound guiding hole(s). Accordingly, various configurations, depending on specific needs, may be obtained by choosing specific position where the sound guiding hole(s) is located, the shape and/or quantity of the sound guiding hole(s) as well as the damping material.

FIG. 5 is a diagram illustrating the equal-loudness contour curves according to some embodiments of the present disclosure. The horizontal coordinate is frequency, while the vertical coordinate is sound pressure level (SPL). As used herein, the SPL refers to the change of atmospheric pressure after being disturbed, i.e., a surplus pressure of the atmospheric pressure, which is equivalent to an atmospheric pressure added to a pressure change caused by the disturbance. As a result, the sound pressure may reflect the amplitude of a sound wave. In FIG. 5, on each curve, sound pressure levels corresponding to different frequencies are different, while the loudness levels felt by human ears are the same. For example, each curve is labeled with a number representing the loudness level of said curve. According to the loudness level curves, when volume (sound pressure amplitude) is lower, human ears are not sensitive to sounds of high or low frequencies; when volume is higher, human ears are more sensitive to sounds of high or low frequencies. Bone conduction speakers may generate sound relating to different frequency ranges, such as 1000 Hz~4000 Hz, or 1000 Hz~4000 Hz, or 1000 Hz~3500 Hz, or 1000 Hz~3000 Hz, or 1500 Hz~3000 Hz. The sound leakage within the

above-mentioned frequency ranges may be the sound leakage aimed to be reduced with a priority.

FIG. 4D is a diagram illustrating the effect of reduced sound leakage according to some embodiments of the present disclosure, wherein the test results and calculation results are close in the above range. The bone conduction speaker being tested includes a cylindrical housing, which includes a sidewall and a bottom, as described in FIGS. 4A and 4B. The cylindrical housing is in a cylinder shape having a radius of 22 mm, the sidewall height of 14 mm, and a plurality of sound guiding holes being set on the upper portion of the sidewall of the housing. The openings of the sound guiding holes are rectangle. The sound guiding holes are arranged evenly on the sidewall. The target region where the sound leakage is to be reduced is 50 cm away from the outside of the bottom of the housing. The distance of the leaked sound wave spreading to the target region and the distance of the sound wave spreading from the surface of the transducer **20** through the sound guiding holes **30** to the target region have a difference of about 180 degrees in phase. As shown, the leaked sound wave is reduced in the target region dramatically or even be eliminated.

According to the embodiments in this disclosure, the effectiveness of reducing sound leakage after setting sound guiding holes is very obvious. As shown in FIG. 4D, the bone conduction speaker having sound guiding holes greatly reduce the sound leakage compared to the bone conduction speaker without sound guiding holes.

In the tested frequency range, after setting sound guiding holes, the sound leakage is reduced by about 10 dB on average. Specifically, in the frequency range of 1500 Hz~3000 Hz, the sound leakage is reduced by over 10 dB. In the frequency range of 2000 Hz~2500 Hz, the sound leakage is reduced by over 20 dB compared to the scheme without sound guiding holes.

A person having ordinary skill in the art can understand from the above-mentioned formulas that when the dimensions of the bone conduction speaker, target regions to reduce sound leakage and frequencies of sound waves differ, the position, shape and quantity of sound guiding holes also need to adjust accordingly.

For example, in a cylinder housing, according to different needs, a plurality of sound guiding holes may be on the sidewall and/or the bottom of the housing. Preferably, the sound guiding hole may be set on the upper portion and/or lower portion of the sidewall of the housing. The quantity of the sound guiding holes set on the sidewall of the housing is no less than two. Preferably, the sound guiding holes may be arranged evenly or unevenly in one or more circles with respect to the center of the bottom. In some embodiments, the sound guiding holes may be arranged in at least one circle. In some embodiments, one sound guiding hole may be set on the bottom of the housing. In some embodiments, the sound guiding hole may be set at the center of the bottom of the housing.

The quantity of the sound guiding holes can be one or more. Preferably, multiple sound guiding holes may be set symmetrically on the housing. In some embodiments, there are 6-8 circularly arranged sound guiding holes.

The openings (and cross sections) of sound guiding holes may be circle, ellipse, rectangle, or slit. Slit generally means slit along with straight lines, curve lines, or arc lines. Different sound guiding holes in one bone conduction speaker may have same or different shapes.

A person having ordinary skill in the art can understand that, the sidewall of the housing may not be cylindrical, the sound guiding holes can be arranged asymmetrically as

11

needed. Various configurations may be obtained by setting different combinations of the shape, quantity, and position of the sound guiding. Some other embodiments along with the figures are described as follows.

In some embodiments, the leaked sound wave may be generated by a portion of the housing **10**. The portion of the housing may be the sidewall **11** of the housing **10** and/or the bottom **12** of the housing **10**. Merely by way of example, the leaked sound wave may be generated by the bottom **12** of the housing **10**. The guided sound wave output through the sound guiding hole(s) **30** may interfere with the leaked sound wave generated by the portion of the housing **10**. The interference may enhance or reduce a sound pressure level of the guided sound wave and/or leaked sound wave in the target region.

In some embodiments, the portion of the housing **10** that generates the leaked sound wave may be regarded as a first sound source (e.g., the sound source **1** illustrated in FIG. **3**), and the sound guiding hole(s) **30** or a part thereof may be regarded as a second sound source (e.g., the sound source **2** illustrated in FIG. **3**). Merely for illustration purposes, if the size of the sound guiding hole on the housing **10** is small, the sound guiding hole may be approximately regarded as a point sound source. In some embodiments, any number or count of sound guiding holes provided on the housing **10** for outputting sound may be approximated as a single point sound source. Similarly, for simplicity, the portion of the housing **10** that generates the leaked sound wave may also be approximately regarded as a point sound source. In some embodiments, both the first sound source and the second sound source may approximately be regarded as point sound sources (also referred to as two-point sound sources).

FIG. **4E** is a schematic diagram illustrating exemplary two-point sound sources according to some embodiments of the present disclosure. The sound field pressure p generated by a single point sound source may satisfy Equation (13):

$$p = \frac{j\omega\rho_0}{4\pi r} Q_0 \exp j(\omega t - kr), \quad (13)$$

where w denotes an angular frequency, ρ_0 denotes an air density, r denotes a distance between a target point and the sound source, Q_0 denotes a volume velocity of the sound source, and k denotes a wave number. It may be concluded that the magnitude of the sound field pressure of the sound field of the point sound source is inversely proportional to the distance to the point sound source.

It should be noted that, the sound guiding hole(s) for outputting sound as a point sound source may only serve as an explanation of the principle and effect of the present disclosure, and the shape and/or size of the sound guiding hole(s) may not be limited in practical applications. In some embodiments, if the area of the sound guiding hole is large, the sound guiding hole may also be equivalent to a planar sound source. Similarly, if an area of the portion of the housing **10** that generates the leaked sound wave is large (e.g., the portion of the housing **10** is a vibration surface or a sound radiation surface), the portion of the housing **10** may also be equivalent to a planar sound source. For those skilled in the art, without creative activities, it may be known that sounds generated by structures such as sound guiding holes, vibration surfaces, and sound radiation surfaces may be equivalent to point sound sources at the spatial scale discussed in the present disclosure, and may have consistent sound propagation characteristics and the same mathemati-

12

cal description method. Further, for those skilled in the art, without creative activities, it may be known that the acoustic effect achieved by the two-point sound sources may also be implemented by alternative acoustic structures. According to actual situations, the alternative acoustic structures may be modified and/or combined discretionarily, and the same acoustic output effect may be achieved.

The two-point sound sources may be formed such that the guided sound wave output from the sound guiding hole(s) may interfere with the leaked sound wave generated by the portion of the housing **10**. The interference may reduce a sound pressure level of the leaked sound wave in the surrounding environment (e.g., the target region). For convenience, the sound waves output from an acoustic output device (e.g., the bone conduction speaker) to the surrounding environment may be referred to as far-field leakage since it may be heard by others in the environment. The sound waves output from the acoustic output device to the ears of the user may also be referred to as near-field sound since a distance between the bone conduction speaker and the user may be relatively short. In some embodiments, the sound waves output from the two-point sound sources may have a same frequency or frequency range (e.g., 800 Hz, 1000 Hz, 1500 Hz, 3000 Hz, etc.). In some embodiments, the sound waves output from the two-point sound sources may have a certain phase difference. In some embodiments, the sound guiding hole includes a damping layer. The damping layer may be, for example, a tuning paper, a tuning cotton, a nonwoven fabric, a silk, a cotton, a sponge, or a rubber. The damping layer may be configured to adjust the phase of the guided sound wave in the target region. The acoustic output device described herein may include a bone conduction speaker or an air conduction speaker. For example, a portion of the housing (e.g., the bottom of the housing) of the bone conduction speaker may be treated as one of the two-point sound sources, and at least one sound guiding holes of the bone conduction speaker may be treated as the other one of the two-point sound sources. As another example, one sound guiding hole of an air conduction speaker may be treated as one of the two-point sound sources, and another sound guiding hole of the air conduction speaker may be treated as the other one of the two-point sound sources. It should be noted that, although the construction of two-point sound sources may be different in bone conduction speaker and air conduction speaker, the principles of the interference between the various constructed two-point sound sources are the same. Thus, the equivalence of the two-point sound sources in a bone conduction speaker disclosed elsewhere in the present disclosure is also applicable for an air conduction speaker.

In some embodiments, when the position and phase difference of the two-point sound sources meet certain conditions, the acoustic output device may output different sound effects in the near field (for example, the position of the user's ear) and the far field. For example, if the phases of the point sound sources corresponding to the portion of the housing **10** and the sound guiding hole(s) are opposite, that is, an absolute value of the phase difference between the two-point sound sources is 180 degrees, the far-field leakage may be reduced according to the principle of reversed phase cancellation.

In some embodiments, the interference between the guided sound wave and the leaked sound wave at a specific frequency may relate to a distance between the sound guiding hole(s) and the portion of the housing **10**. For example, if the sound guiding hole(s) are set at the upper portion of the sidewall of the housing **10** (as illustrated in

13

FIG. 4A), the distance between the sound guiding hole(s) and the portion of the housing 10 may be large. Correspondingly, the frequencies of sound waves generated by such two-point sound sources may be in a mid-low frequency range (e.g., 1500-2000 Hz, 1500-2500 Hz, etc.). Referring to FIG. 4D, the interference may reduce the sound pressure level of the leaked sound wave in the mid-low frequency range (i.e., the sound leakage is low).

Merely by way of example, the low frequency range may refer to frequencies in a range below a first frequency threshold. The high frequency range may refer to frequencies in a range exceed a second frequency threshold. The first frequency threshold may be lower than the second frequency threshold. The mid-low frequency range may refer to frequencies in a range between the first frequency threshold and the second frequency threshold. For example, the first frequency threshold may be 1000 Hz, and the second frequency threshold may be 3000 Hz. The low frequency range may refer to frequencies in a range below 1000 Hz, the high frequency range may refer to frequencies in a range above 3000 Hz, and the mid-low frequency range may refer to frequencies in a range of 1000-2000 Hz, 1500-2500 Hz, etc. In some embodiments, a middle frequency range, a mid-high frequency range may also be determined between the first frequency threshold and the second frequency threshold. In some embodiments, the mid-low frequency range and the low frequency range may partially overlap. The mid-high frequency range and the high frequency range may partially overlap. For example, the mid-high frequency range may refer to frequencies in a range above 3000 Hz, and the mid-low frequency range may refer to frequencies in a range of 2800-3500 Hz. It should be noted that the low frequency range, the mid-low frequency range, the middle frequency range, the mid-high frequency range, and/or the high frequency range may be set flexibly according to different situations, and are not limited herein.

In some embodiments, the frequencies of the guided sound wave and the leaked sound wave may be set in a low frequency range (e.g., below 800 Hz, below 1200 Hz, etc.). In some embodiments, the amplitudes of the sound waves generated by the two-point sound sources may be set to be different in the low frequency range. For example, the amplitude of the guided sound wave may be smaller than the amplitude of the leaked sound wave. In this case, the interference may not reduce sound pressure of the near-field sound in the low-frequency range. The sound pressure of the near-field sound may be improved in the low-frequency range. The volume of the sound heard by the user may be improved.

In some embodiments, the amplitude of the guided sound wave may be adjusted by setting an acoustic resistance structure in the sound guiding hole(s) 30. The material of the acoustic resistance structure disposed in the sound guiding hole 30 may include, but not limited to, plastics (e.g., high-molecular polyethylene, blown nylon, engineering plastics, etc.), cotton, nylon, fiber (e.g., glass fiber, carbon fiber, boron fiber, graphite fiber, graphene fiber, silicon carbide fiber, or aramid fiber), other single or composite materials, other organic and/or inorganic materials, etc. The thickness of the acoustic resistance structure may be 0.005 mm, 0.01 mm, 0.02 mm, 0.5 mm, 1 mm, 2 mm, etc. The structure of the acoustic resistance structure may be in a shape adapted to the shape of the sound guiding hole. For example, the acoustic resistance structure may have a shape of a cylinder, a sphere, a cubic, etc. In some embodiments, the materials, thickness, and structures of the acoustic resistance structure may be modified and/or combined to obtain

14

a desirable acoustic resistance structure. In some embodiments, the acoustic resistance structure may be implemented by the damping layer.

In some embodiments, the amplitude of the guided sound wave output from the sound guiding hole may be relatively low (e.g., zero or almost zero). The difference between the guided sound wave and the leaked sound wave may be maximized, thus achieving a relatively large sound pressure in the near field. In this case, the sound leakage of the acoustic output device having sound guiding holes may be almost the same as the sound leakage of the acoustic output device without sound guiding holes in the low frequency range (e.g., as shown in FIG. 4D).

Embodiment Two

FIG. 6 is a flowchart of an exemplary method of reducing sound leakage of a bone conduction speaker according to some embodiments of the present disclosure. At 601, a bone conduction speaker including a vibration plate 21 touching human skin and passing vibrations, a transducer 22, and a housing 10 is provided. At least one sound guiding hole 30 is arranged on the housing 10. At 602, the vibration plate 21 is driven by the transducer 22, causing the vibration 21 to vibrate. At 603, a leaked sound wave due to the vibrations of the housing is formed, wherein the leaked sound wave transmits in the air. At 604, a guided sound wave passing through the at least one sound guiding hole 30 from the inside to the outside of the housing 10. The guided sound wave interferes with the leaked sound wave, reducing the sound leakage of the bone conduction speaker.

The sound guiding holes 30 are preferably set at different positions of the housing 10.

The effectiveness of reducing sound leakage may be determined by the formulas and method as described above, based on which the positions of sound guiding holes may be determined.

A damping layer is preferably set in a sound guiding hole 30 to adjust the phase and amplitude of the sound wave transmitted through the sound guiding hole 30.

In some embodiments, different sound guiding holes may generate different sound waves having a same phase to reduce the leaked sound wave having the same wavelength. In some embodiments, different sound guiding holes may generate different sound waves having different phases to reduce the leaked sound waves having different wavelengths.

In some embodiments, different portions of a sound guiding hole 30 may be configured to generate sound waves having a same phase to reduce the leaked sound waves with the same wavelength. In some embodiments, different portions of a sound guiding hole 30 may be configured to generate sound waves having different phases to reduce the leaked sound waves with different wavelengths.

Additionally, the sound wave inside the housing may be processed to basically have the same value but opposite phases with the leaked sound wave, so that the sound leakage may be further reduced.

Embodiment Three

FIGS. 7A and 7B are schematic structures illustrating an exemplary bone conduction speaker according to some embodiments of the present disclosure. The bone conduction speaker may include an open housing 10, a vibration board 21, and a transducer 22. The housing 10 may cylindrical and have a sidewall and a bottom. A plurality of sound guiding

15

holes **30** may be arranged on the lower portion of the sidewall (i.e., from about the $\frac{2}{3}$ height of the sidewall to the bottom). The quantity of the sound guiding holes **30** may be 8, the openings of the sound guiding holes **30** may be rectangle. The sound guiding holes **30** may be arranged evenly or unevenly in one or more circles on the sidewall of the housing **10**.

In the embodiment, the transducer **22** is preferably implemented based on the principle of electromagnetic transduction. The transducer may include components such as magnetizer, voice coil, and etc., and the components may locate inside the housing and may generate synchronous vibrations with a same frequency.

FIG. 7C is a diagram illustrating reduced sound leakage according to some embodiments of the present disclosure. In the frequency range of 1400 Hz~4000 Hz, the sound leakage is reduced by more than 5 dB, and in the frequency range of 2250 Hz~2500 Hz, the sound leakage is reduced by more than 20 dB.

In some embodiments, the sound guiding hole(s) at the lower portion of the sidewall of the housing **10** may also be approximately regarded as a point sound source. In some embodiments, the sound guiding hole(s) at the lower portion of the sidewall of the housing **10** and the portion of the housing **10** that generates the leaked sound wave may constitute two-point sound sources. The two-point sound sources may be formed such that the guided sound wave output from the sound guiding hole(s) at the lower portion of the sidewall of the housing **10** may interfere with the leaked sound wave generated by the portion of the housing **10**. The interference may reduce a sound pressure level of the leaked sound wave in the surrounding environment (e.g., the target region) at a specific frequency or frequency range.

In some embodiments, the sound waves output from the two-point sound sources may have a same frequency or frequency range (e.g., 1000 Hz, 2500 Hz, 3000 Hz, etc.). In some embodiments, the sound waves output from the first two-point sound sources may have a certain phase difference. In this case, the interference between the sound waves generated by the first two-point sound sources may reduce a sound pressure level of the leaked sound wave in the target region. When the position and phase difference of the first two-point sound sources meet certain conditions, the acoustic output device may output different sound effects in the near field (for example, the position of the user's ear) and the far field. For example, if the phases of the first two-point sound sources are opposite, that is, an absolute value of the phase difference between the first two-point sound sources is 180 degrees, the far-field leakage may be reduced.

In some embodiments, the interference between the guided sound wave and the leaked sound wave may relate to frequencies of the guided sound wave and the leaked sound wave and/or a distance between the sound guiding hole(s) and the portion of the housing **10**. For example, if the sound guiding hole(s) are set at the lower portion of the sidewall of the housing **10** (as illustrated in FIG. 7A), the distance between the sound guiding hole(s) and the portion of the housing **10** may be small. Correspondingly, the frequencies of sound waves generated by such two-point sound sources may be in a high frequency range (e.g., above 3000 Hz, above 3500 Hz, etc.). Referring to FIG. 7C, the interference may reduce the sound pressure level of the leaked sound wave in the high frequency range.

Embodiment Four

FIGS. 8A and 8B are schematic structures illustrating an exemplary bone conduction speaker according to some

16

embodiments of the present disclosure. The bone conduction speaker may include an open housing **10**, a vibration board **21**, and a transducer **22**. The housing **10** is cylindrical and have a sidewall and a bottom. The sound guiding holes **30** may be arranged on the central portion of the sidewall of the housing (i.e., from about the $\frac{1}{3}$ height of the sidewall to the $\frac{2}{3}$ height of the sidewall). The quantity of the sound guiding holes **30** may be 8, and the openings (and cross sections) of the sound guiding hole **30** may be rectangle. The sound guiding holes **30** may be arranged evenly or unevenly in one or more circles on the sidewall of the housing **10**.

In the embodiment, the transducer **21** may be implemented preferably based on the principle of electromagnetic transduction. The transducer **21** may include components such as magnetizer, voice coil, etc., which may be placed inside the housing and may generate synchronous vibrations with the same frequency.

FIG. 8C is a diagram illustrating reduced sound leakage. In the frequency range of 1000 Hz~4000 Hz, the effectiveness of reducing sound leakage is great. For example, in the frequency range of 1400 Hz~2900 Hz, the sound leakage is reduced by more than 10 dB; in the frequency range of 2200 Hz~2500 Hz, the sound leakage is reduced by more than 20 dB.

It's illustrated that the effectiveness of reduced sound leakage can be adjusted by changing the positions of the sound guiding holes, while keeping other parameters relating to the sound guiding holes unchanged.

Embodiment Five

FIGS. 9A and 9B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure. The bone conduction speaker may include an open housing **10**, a vibration board **21** and a transducer **22**. The housing **10** is cylindrical, with a sidewall and a bottom. One or more perforative sound guiding holes **30** may be along the circumference of the bottom. In some embodiments, there may be 8 sound guiding holes **30** arranged evenly or unevenly in one or more circles on the bottom of the housing **10**. In some embodiments, the shape of one or more of the sound guiding holes **30** may be rectangle.

In the embodiment, the transducer **21** may be implemented preferably based on the principle of electromagnetic transduction. The transducer **21** may include components such as magnetizer, voice coil, etc., which may be placed inside the housing and may generate synchronous vibration with the same frequency.

FIG. 9C is a diagram illustrating the effect of reduced sound leakage. In the frequency range of 1000 Hz~3000 Hz, the effectiveness of reducing sound leakage is outstanding. For example, in the frequency range of 1700 Hz~2700 Hz, the sound leakage is reduced by more than 10 dB; in the frequency range of 2200 Hz~2400 Hz, the sound leakage is reduced by more than 20 dB.

Embodiment Six

FIGS. 10A and 10B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure. The bone conduction speaker may include an open housing **10**, a vibration board **21** and a transducer **22**. One or more perforative sound guiding holes **30** may be arranged on both upper and lower portions of the sidewall of the housing **10**. The sound guiding holes **30** may be arranged evenly or unevenly in one

or more circles on the upper and lower portions of the sidewall of the housing 10. In some embodiments, the quantity of sound guiding holes 30 in every circle may be 8, and the upper portion sound guiding holes and the lower portion sound guiding holes may be symmetrical about the central cross section of the housing 10. In some embodiments, the shape of the sound guiding hole 30 may be circle.

The shape of the sound guiding holes on the upper portion and the shape of the sound guiding holes on the lower portion may be different; One or more damping layers may be arranged in the sound guiding holes to reduce leaked sound waves of the same wave length (or frequency), or to reduce leaked sound waves of different wave lengths.

FIG. 10C is a diagram illustrating the effect of reducing sound leakage according to some embodiments of the present disclosure. In the frequency range of 1000 Hz~4000 Hz, the effectiveness of reducing sound leakage is outstanding. For example, in the frequency range of 1600 Hz~2700 Hz, the sound leakage is reduced by more than 15 dB; in the frequency range of 2000 Hz~2500 Hz, where the effectiveness of reducing sound leakage is most outstanding, the sound leakage is reduced by more than 20 dB. Compared to embodiment three, this scheme has a relatively balanced effect of reduced sound leakage on various frequency range, and this effect is better than the effect of schemes where the height of the holes are fixed, such as schemes of embodiment three, embodiment four, embodiment five, and so on.

In some embodiments, the sound guiding hole(s) at the upper portion of the sidewall of the housing 10 (also referred to as first hole(s)) may be approximately regarded as a point sound source. In some embodiments, the first hole(s) and the portion of the housing 10 that generates the leaked sound wave may constitute two-point sound sources (also referred to as first two-point sound sources). As for the first two-point sound sources, the guided sound wave generated by the first hole(s) (also referred to as first guided sound wave) may interfere with the leaked sound wave or a portion thereof generated by the portion of the housing 10 in a first region. In some embodiments, the sound waves output from the first two-point sound sources may have a same frequency (e.g., a first frequency). In some embodiments, the sound waves output from the first two-point sound sources may have a certain phase difference. In this case, the interference between the sound waves generated by the first two-point sound sources may reduce a sound pressure level of the leaked sound wave in the target region. When the position and phase difference of the first two-point sound sources meet certain conditions, the acoustic output device may output different sound effects in the near field (for example, the position of the user's ear) and the far field. For example, if the phases of the first two-point sound sources are opposite, that is, an absolute value of the phase difference between the first two-point sound sources is 180 degrees, the far-field leakage may be reduced according to the principle of reversed phase cancellation.

In some embodiments, the sound guiding hole(s) at the lower portion of the sidewall of the housing 10 (also referred to as second hole(s)) may also be approximately regarded as another point sound source. Similarly, the second hole(s) and the portion of the housing 10 that generates the leaked sound wave may also constitute two-point sound sources (also referred to as second two-point sound sources). As for the second two-point sound sources, the guided sound wave generated by the second hole(s) (also referred to as second guided sound wave) may interfere with the leaked sound wave or a portion thereof generated by the portion of the housing 10 in a second region. The second region may be the

same as or different from the first region. In some embodiments, the sound waves output from the second two-point sound sources may have a same frequency (e.g., a second frequency).

In some embodiments, the first frequency and the second frequency may be in certain frequency ranges. In some embodiments, the frequency of the guided sound wave output from the sound guiding hole(s) may be adjustable. In some embodiments, the frequency of the first guided sound wave and/or the second guided sound wave may be adjusted by one or more acoustic routes. The acoustic routes may be coupled to the first hole(s) and/or the second hole(s). The first guided sound wave and/or the second guided sound wave may be propagated along the acoustic route having a specific frequency selection characteristic. That is, the first guided sound wave and the second guided sound wave may be transmitted to their corresponding sound guiding holes via different acoustic routes. For example, the first guided sound wave and/or the second guided sound wave may be propagated along an acoustic route with a low-pass characteristic to a corresponding sound guiding hole to output guided sound wave of a low frequency. In this process, the high frequency component of the sound wave may be absorbed or attenuated by the acoustic route with the low-pass characteristic. Similarly, the first guided sound wave and/or the second guided sound wave may be propagated along an acoustic route with a high-pass characteristic to the corresponding sound guiding hole to output guided sound wave of a high frequency. In this process, the low frequency component of the sound wave may be absorbed or attenuated by the acoustic route with the high-pass characteristic.

FIG. 10D is a schematic diagram illustrating an acoustic route according to some embodiments of the present disclosure. FIG. 10E is a schematic diagram illustrating another acoustic route according to some embodiments of the present disclosure. FIG. 10F is a schematic diagram illustrating a further acoustic route according to some embodiments of the present disclosure. In some embodiments, structures such as a sound tube, a sound cavity, a sound resistance, etc., may be set in the acoustic route for adjusting frequencies for the sound waves (e.g., by filtering certain frequencies). It should be noted that FIGS. 10D-10F may be provided as examples of the acoustic routes, and not intended be limiting.

As shown in FIG. 10D, the acoustic route may include one or more lumen structures. The one or more lumen structures may be connected in series. An acoustic resistance material may be provided in each of at least one of the one or more lumen structures to adjust acoustic impedance of the entire structure to achieve a desirable sound filtering effect. For example, the acoustic impedance may be in a range of 5 MKS Rayleigh to 500 MKS Rayleigh. In some embodiments, a high-pass sound filtering, a low-pass sound filtering, and/or a band-pass filtering effect of the acoustic route may be achieved by adjusting a size of each of at least one of the one or more lumen structures and/or a type of acoustic resistance material in each of at least one of the one or more lumen structures. The acoustic resistance materials may include, but not limited to, plastic, textile, metal, permeable material, woven material, screen material or mesh material, porous material, particulate material, polymer material, or the like, or any combination thereof. By setting the acoustic routes of different acoustic impedances, the acoustic output from the sound guiding holes may be acoustically filtered. In this case, the guided sound waves may have different frequency components.

19

As shown in FIG. 10E, the acoustic route may include one or more resonance cavities. The one or more resonance cavities may be, for example, Helmholtz cavity. In some embodiments, a high-pass sound filtering, a low-pass sound filtering, and/or a band-pass filtering effect of the acoustic route may be achieved by adjusting a size of each of at least one of the one or more resonance cavities and/or a type of acoustic resistance material in each of at least one of the one or more resonance cavities.

As shown in FIG. 10F, the acoustic route may include a combination of one or more lumen structures and one or more resonance cavities. In some embodiments, a high-pass sound filtering, a low-pass sound filtering, and/or a band-pass filtering effect of the acoustic route may be achieved by adjusting a size of each of at least one of the one or more lumen structures and one or more resonance cavities and/or a type of acoustic resistance material in each of at least one of the one or more lumen structures and one or more resonance cavities. It should be noted that the structures exemplified above may be for illustration purposes, various acoustic structures may also be provided, such as a tuning net, tuning cotton, etc.

In some embodiments, the interference between the leaked sound wave and the guided sound wave may relate to frequencies of the guided sound wave and the leaked sound wave and/or a distance between the sound guiding hole(s) and the portion of the housing 10. In some embodiments, the portion of the housing that generates the leaked sound wave may be the bottom of the housing 10. The first hole(s) may have a larger distance to the portion of the housing 10 than the second hole(s). In some embodiments, the frequency of the first guided sound wave output from the first hole(s) (e.g., the first frequency) and the frequency of second guided sound wave output from second hole(s) (e.g., the second frequency) may be different.

In some embodiments, the first frequency and second frequency may associate with the distance between the at least one sound guiding hole and the portion of the housing 10 that generates the leaked sound wave. In some embodiments, the first frequency may be set in a low frequency range. The second frequency may be set in a high frequency range. The low frequency range and the high frequency range may or may not overlap.

In some embodiments, the frequency of the leaked sound wave generated by the portion of the housing 10 may be in a wide frequency range. The wide frequency range may include, for example, the low frequency range and the high frequency range or a portion of the low frequency range and the high frequency range. For example, the leaked sound wave may include a first frequency in the low frequency range and a second frequency in the high frequency range. In some embodiments, the leaked sound wave of the first frequency and the leaked sound wave of the second frequency may be generated by different portions of the housing 10. For example, the leaked sound wave of the first frequency may be generated by the sidewall of the housing 10, the leaked sound wave of the second frequency may be generated by the bottom of the housing 10. As another example, the leaked sound wave of the first frequency may be generated by the bottom of the housing 10, the leaked sound wave of the second frequency may be generated by the sidewall of the housing 10. In some embodiments, the frequency of the leaked sound wave generated by the portion of the housing 10 may relate to parameters including the mass, the damping, the stiffness, etc., of the different portion of the housing 10, the frequency of the transducer 22, etc.

20

In some embodiments, the characteristics (amplitude, frequency, and phase) of the first two-point sound sources and the second two-point sound sources may be adjusted via various parameters of the acoustic output device (e.g., electrical parameters of the transducer 22, the mass, stiffness, size, structure, material, etc., of the portion of the housing 10, the position, shape, structure, and/or number (or count) of the sound guiding hole(s) so as to form a sound field with a particular spatial distribution. In some embodiments, a frequency of the first guided sound wave is smaller than a frequency of the second guided sound wave.

A combination of the first two-point sound sources and the second two-point sound sources may improve sound effects both in the near field and the far field.

Referring to FIGS. 4D, 7C, and 10C, by designing different two-point sound sources with different distances, the sound leakage in both the low frequency range and the high frequency range may be properly suppressed. In some embodiments, the closer distance between the second two-point sound sources may be more suitable for suppressing the sound leakage in the far field, and the relative longer distance between the first two-point sound sources may be more suitable for reducing the sound leakage in the near field. In some embodiments, the amplitudes of the sound waves generated by the first two-point sound sources may be set to be different in the low frequency range. For example, the amplitude of the guided sound wave may be smaller than the amplitude of the leaked sound wave. In this case, the sound pressure level of the near-field sound may be improved. The volume of the sound heard by the user may be increased.

Embodiment Seven

FIGS. 11A and 11B are schematic structures illustrating a bone conduction speaker according to some embodiments of the present disclosure. The bone conduction speaker may include an open housing 10, a vibration board 21 and a transducer 22. One or more perforative sound guiding holes 30 may be set on upper and lower portions of the sidewall of the housing 10 and on the bottom of the housing 10. The sound guiding holes 30 on the sidewall are arranged evenly or unevenly in one or more circles on the upper and lower portions of the sidewall of the housing 10. In some embodiments, the quantity of sound guiding holes 30 in every circle may be 8, and the upper portion sound guiding holes and the lower portion sound guiding holes may be symmetrical about the central cross section of the housing 10. In some embodiments, the shape of the sound guiding hole 30 may be rectangular. There may be four sound guiding holes 30 on the bottom of the housing 10. The four sound guiding holes 30 may be linear-shaped along arcs, and may be arranged evenly or unevenly in one or more circles with respect to the center of the bottom. Furthermore, the sound guiding holes 30 may include a circular perforative hole on the center of the bottom.

FIG. 11C is a diagram illustrating the effect of reducing sound leakage of the embodiment. In the frequency range of 1000 Hz~4000 Hz, the effectiveness of reducing sound leakage is outstanding. For example, in the frequency range of 1300 Hz~3000 Hz, the sound leakage is reduced by more than 10 dB; in the frequency range of 2000 Hz~2700 Hz, the sound leakage is reduced by more than 20 dB. Compared to embodiment three, this scheme has a relatively balanced effect of reduced sound leakage within various frequency range, and this effect is better than the effect of schemes where the height of the holes are fixed, such as schemes of

21

embodiment three, embodiment four, embodiment five, and etc. Compared to embodiment six, in the frequency range of 1000 Hz~1700 Hz and 2500 Hz~4000 Hz, this scheme has a better effect of reduced sound leakage than embodiment six.

Embodiment Eight

FIGS. 12A and 12B are schematic structures illustrating a bone conduction speaker according to some embodiments of the present disclosure. The bone conduction speaker may include an open housing 10, a vibration board 21 and a transducer 22. A perforative sound guiding hole 30 may be set on the upper portion of the sidewall of the housing 10. One or more sound guiding holes may be arranged evenly or unevenly in one or more circles on the upper portion of the sidewall of the housing 10. There may be 8 sound guiding holes 30, and the shape of the sound guiding holes 30 may be circle.

After comparison of calculation results and test results, the effectiveness of this embodiment is basically the same with that of embodiment one, and this embodiment can effectively reduce sound leakage.

Embodiment Nine

FIGS. 13A and 13B are schematic structures illustrating a bone conduction speaker according to some embodiments of the present disclosure. The bone conduction speaker may include an open housing 10, a vibration board 21 and a transducer 22.

The difference between this embodiment and the above-described embodiment three is that to reduce sound leakage to greater extent, the sound guiding holes 30 may be arranged on the upper, central and lower portions of the sidewall 11. The sound guiding holes 30 are arranged evenly or unevenly in one or more circles. Different circles are formed by the sound guiding holes 30, one of which is set along the circumference of the bottom 12 of the housing 10. The size of the sound guiding holes 30 are the same.

The effect of this scheme may cause a relatively balanced effect of reducing sound leakage in various frequency ranges compared to the schemes where the position of the holes are fixed. The effect of this design on reducing sound leakage is relatively better than that of other designs where the heights of the holes are fixed, such as embodiment three, embodiment four, embodiment five, etc.

Embodiment Ten

The sound guiding holes 30 in the above embodiments may be perforative holes without shields.

In order to adjust the effect of the sound waves guided from the sound guiding holes, a damping layer (not shown in the figures) may locate at the opening of a sound guiding hole 30 to adjust the phase and/or the amplitude of the sound wave.

There are multiple variations of materials and positions of the damping layer. For example, the damping layer may be made of materials which can damp sound waves, such as tuning paper, tuning cotton, nonwoven fabric, silk, cotton, sponge or rubber. The damping layer may be attached on the inner wall of the sound guiding hole 30, or may shield the sound guiding hole 30 from outside.

More preferably, the damping layers corresponding to different sound guiding holes 30 may be arranged to adjust the sound waves from different sound guiding holes to

22

generate a same phase. The adjusted sound waves may be used to reduce leaked sound wave having the same wavelength. Alternatively, different sound guiding holes 30 may be arranged to generate different phases to reduce leaked sound wave having different wavelengths (i.e., leaked sound waves with specific wavelengths).

In some embodiments, different portions of a same sound guiding hole can be configured to generate a same phase to reduce leaked sound waves on the same wavelength (e.g., using a pre-set damping layer with the shape of stairs or steps). In some embodiments, different portions of a same sound guiding hole can be configured to generate different phases to reduce leaked sound waves on different wavelengths.

The above-described embodiments are preferable embodiments with various configurations of the sound guiding hole(s) on the housing of a bone conduction speaker, but a person having ordinary skills in the art can understand that the embodiments don't limit the configurations of the sound guiding hole(s) to those described in this application.

In the past bone conduction speakers, the housing of the bone conduction speakers is closed, so the sound source inside the housing is sealed inside the housing. In the embodiments of the present disclosure, there can be holes in proper positions of the housing, making the sound waves inside the housing and the leaked sound waves having substantially same amplitude and substantially opposite phases in the space, so that the sound waves can interfere with each other and the sound leakage of the bone conduction speaker is reduced. Meanwhile, the volume and weight of the speaker do not increase, the reliability of the product is not comprised, and the cost is barely increased. The designs disclosed herein are easy to implement, reliable, and effective in reducing sound leakage.

In some embodiments, an environmental sound collection and processing function may be added to a speaker as described elsewhere in the present disclosure, e.g., to enable the speaker to implement the function of a hearing aid, or to collect the voice of the user/wearer to enable voice communication with others. For example, an electronic component including a microphone may be added to the speaker. The microphone may collect environmental sounds of a user/wearer, process the sounds using an algorithm and transmit the processed sound (or generated electrical signal) to the user/wearer of the speaker. That is, the speaker may be modified to include the function of collecting the environmental sounds, and after a signal processing, the sound may be transmitted to the user/wearer via the speaker, thereby implementing the function of the hearing aid. The algorithm mentioned herein may include noise cancellation, automatic gain control, acoustic feedback suppression, wide dynamic range compression, active environment recognition, active noise reduction, directional processing, tinnitus processing, multi-channel wide dynamic range compression, active howling suppression, volume control, or the like, or any combination thereof.

FIG. 14 is a sectional view illustrating an electronic component according to the present disclosure. The electronic component may be a portion of a speaker described elsewhere in the present disclosure. As shown in FIG. 14, the electronic component may include a first microphone element 1412, a bracket 141, a circuit component (e.g., including a first circuit board 142-1, a second circuit board 142-2), a cover layer 143, a chamber 145, etc. As used herein, the bracket 141 may be used to physically connect to an accommodation body of the speaker. The cover layer 143 may integrally form on the surface of the bracket 141 by

injection molding to provide a seal for the chamber 145 after the bracket 141 is connected to the accommodation body. In some embodiments, the first microphone element 1412 may be disposed on the first circuit board 142-1 of the circuit component to be accommodated inside the chamber 145. The first microphone element 1412 may be used to receive a sound signal from the outside of the electronic component, and convert the sound signal into an electrical signal for analysis and processing. In some embodiments, the first microphone element 1412 may also be referred to as a microphone 1412 for brevity.

In some embodiments, the bracket 141 may be disposed with a microphone hole corresponding to the first microphone element 1412. The cover layer 143 may be disposed with a first sound guiding hole 1223 corresponding to the microphone hole. A first sound blocking member 1224 may be disposed at a position corresponding to the microphone hole. The first sound blocking member 1224 may extend towards the inside of the chamber 145 via the microphone hole and define a sound guiding channel 12241. One end of the sound guiding channel 12241 may be in communication with the first sound guiding hole 1223 of the cover layer 143. The first microphone element 1412 may be inserted into the sound guiding channel 12241 from another end of the sound guiding channel 12241.

In some embodiments, the electronic component may also include a switch in the embodiment. The circuit component may be disposed with the switch. The switch may be disposed on an outer side of the first circuit board 142-1 towards an opening of the chamber 145. Correspondingly, the bracket 141 may be disposed with a switch hole corresponding to the switch. The cover layer 143 may further cover the switch hole. The switch hole and the microphone hole may be disposed on the bracket 141 at intervals.

In some embodiments, the first sound guiding hole 1223 may be disposed through the cover layer 143 and correspond to the position of the first microphone element 1412. The first sound guiding hole 1223 may correspond to the microphone hole of the bracket 141, and further communicate the first microphone element 1412 with the outside of the electronic component. Therefore, a sound from the outside of the electronic component may be received by the first microphone element 1412 via the first sound guiding hole 1223 and the microphone hole.

The shape of the first sound guiding hole 1223 may be any shape as long as the sound from the outside of the electronic component is able to be received by the electronic component. In some embodiments, the first sound guiding hole 1223 may be a circular hole having a relatively small size, and disposed in a region of the cover layer 143 corresponding to the microphone hole. The first sound guiding hole 1223 with the small size may limit the communication between the first microphone element 1412 or the like in the electronic component and the outside, thereby improving the sealing of the electronic component.

In some embodiments, the first sound blocking member 1224 may extend to the periphery of the first microphone element 1412 from the cover layer 143, through the periphery of the first sound guiding hole 1223, the microphone hole and the inside of the chamber 145 to form the sound guiding channel 12241 from the first sound guiding hole 1223 to the first microphone element 1412. Therefore, the sound signal of the electronic component entering the sound guiding hole may directly reach the first microphone element 1412 through the sound guiding channel 12241.

In some embodiments, a shape of the sound guiding channel 12241 in a section perpendicular to the length

direction may be the same as or different from the shape of the microphone hole or the first microphone element 1412. In some embodiments, the sectional shapes of the microphone hole and the first microphone element 1412 in a direction perpendicular to the bracket 141 towards the chamber 145 may be square. The size of the microphone hole may be slightly larger than the outer size of the sound guiding channel 12241. The inner size of the sound guiding channel 12241 may not be less than the outer size of the first microphone element 1412. Therefore, the sound guiding channel 12241 may pass through the first sound guiding hole 1223 to reach the first microphone element 1412 and be wrapped around the periphery of the first microphone element 1412.

Through the way described above, the cover layer 143 of the electronic component may be disposed with the first sound guiding hole 1223 and the sound guiding channel 12241 passing from the periphery of the first sound guiding hole 1223 through the microphone hole to reach the first microphone element 1412 and wrapped around the periphery of the first microphone element 1412. The sound guiding channel 12241 may be disposed so that the sound signal entering through the first sound guiding hole 1223 may reach the first microphone element 1412 via the first sound guiding hole 1223 and be received by the first microphone element 1412. Therefore, the leakage of the sound signal in the transmission process may be reduced, thereby improving the efficiency of receiving the electronic signal by the electronic component.

In some embodiments, the electronic component may also include a waterproof mesh cloth 146 disposed inside the sound guiding channel 12241. The waterproof mesh cloth 146 may be held against the side of the cover layer 143 towards the microphone element by the first microphone element 1412 and cover the first sound guiding hole 1223.

In some embodiments, the bracket 141 may protrude at a position of the bracket 141 close to the first microphone element 1412 in the sound guiding channel 12241 to form a convex surface opposite to the first microphone element 1412. Therefore, the waterproof mesh cloth 146 may be sandwiched between the first microphone element 1412 and the convex surface, or directly adhered to the periphery of the first microphone element 1412, and the specific setting manner may not be limited herein.

In addition to the waterproof function for the first microphone element 1412, the waterproof mesh cloth 146 in the embodiment may also have a function of sound transmission, etc., to avoid adversely affecting the sound receiving effect of a sound receiving region 13121 of the first microphone element 1412.

In some embodiments, the cover layer 143 may be arranged in a stripe shape. As used herein, a main axis of the first sound guiding hole 1223 and a main axis of the sound receiving region 13121 of the first microphone element 1412 may be spaced from each other in the width direction of the cover layer 143. As used herein, the main axis of the sound receiving region 13121 of the first microphone element 1412 may refer to a main axis of the sound receiving region 13121 of the first microphone element 1412 in the width direction of the cover layer 143, such as an axis n in FIG. 14. The main axis of the first sound guiding hole 1223 may be an axis m in FIG. 14.

It should be noted that, due to the setting requirements of the circuit component, the first microphone element 1412 may be disposed at a first position of the first circuit board 142-1. When the first sound guiding hole 1223 is disposed, the first sound guiding hole 1223 may be disposed at a

25

second position of the cover layer **143** due to the aesthetic and convenient requirements. In the embodiment, the first position and the second position may not correspond in the width direction of the cover layer **143**. Therefore, the main axis of the first sound guiding hole **1223** and the main axis of the sound receiving region **13121** of the first microphone element **1412** may be spaced from each other in the width direction of the cover layer **143**. Therefore, the sound input via the first sound guiding hole **1223** may not reach the sound receiving region **13121** of the first microphone element **1412** along a straight line.

In some embodiments, in order to guide the sound signal entered from the first sound guiding hole **1223** to the first microphone element **1412**, the sound guiding channel **12241** may be disposed with a curved shape.

In some embodiments, the main axis of the first sound guiding hole **1223** may be disposed in the middle of the cover layer **143** in the width direction of the cover layer **143**.

In some embodiments, the cover layer **143** may be a portion of the outer housing of the electronic device. In order to meet the overall aesthetic requirements of the electronic device, the first sound guiding hole **1223** may be disposed in the middle in the width direction of the cover layer **143**. Therefore, the first sound guiding hole **1223** may look more symmetrical and meet the visual requirements of people.

In some embodiments, the corresponding sound guiding channel **12241** may be disposed with a stepped shape in a section. Therefore, the sound signal introduced by the first sound guiding hole **1223** may be transmitted to the first microphone element **1412** through the stepped sound guiding channel **12241** and received by the first microphone element **1412**.

FIG. **15** is a partial structural diagram illustrating a speaker according to an embodiment of the present disclosure. FIG. **16** is an exploded diagram illustrating a partial structure of a speaker according to an embodiment of the present disclosure. FIG. **17** is a sectional view illustrating a partial structure of a speaker according to an embodiment of the present disclosure. The speaker described herein may be similar to a speaker described elsewhere in the present disclosure. It should be noted that, without departing from the spirit and scope of the present disclosure, the contents described below may be applied to an air conduction speaker and a bone conduction speaker.

Referring to FIG. **15** and FIG. **16**, in some embodiments, the speaker may include one or more microphones. The number (or count) of the microphones may include two, i.e., a first microphone **1532a** and a second microphone **1532b**. As used herein, the first microphone **1532a** and the second microphone **1532b** may both be MEMS (micro-electromechanical system) microphones which may have a small working current, relatively stable performance, and high voice quality. The two microphones may be disposed at different positions of a flexible circuit board **154** according to actual requirements.

In some embodiments, the flexible circuit board **154** may be disposed in the speaker. The flexible circuit board **154** may include a main circuit board **1541**, and a branch circuit board **1542** and a branch circuit board **1543** connected to the main circuit board **1541**. The branch circuit board **1542** may extend in the same direction as the main circuit board **1541**. The first microphone **1532a** may be disposed on one end of the branch circuit board **1542** away from the main circuit board **1541**. The branch circuit board **1543** may extend perpendicular to the main circuit board **1541**. The second microphone **1532b** may be disposed on one end of the branch circuit board **1543** away from the main circuit board

26

1541. A plurality of pads **155** may be disposed on the end of the main circuit board **1541** away from the branch circuit board **1542** and the branch circuit board **1543**. The one or more microphones may be connected to the main circuit board **1541** by one or more wires (e.g., a wire **157**, a wire **159**, etc.).

In some embodiments, a core housing (which is the same as or similar to the housing **10** as described in FIGS. **1-13**) may include a peripheral side wall **1511** and a bottom end wall **1512** connected to one end surface of the peripheral side wall **1511** to form an accommodation space with an open end. As used herein, an earphone core may be placed in the accommodation space through the open end. The first microphone **1532a** may be fixed on the bottom end wall **1512**. The second microphone **1532b** may be fixed on the peripheral side wall **1511**.

In some embodiments, the branch circuit board **1542** and/or the branch circuit board **1543** may be appropriately bent to suit a position of a sound inlet corresponding to the microphone at the core housing. Specifically, the flexible circuit board **154** may be disposed in the core housing in a manner that the main circuit board **1541** is parallel to the bottom end wall **1512**. Therefore, the first microphone **1532a** may correspond to the bottom end wall **1512** without bending the main circuit board **1541**. Since the second microphone **1532b** may be fixed to the peripheral side wall **1511** of the core housing, it may be necessary to bend the main circuit board **1541**. Specifically, the branch circuit board **1543** may be bent at one end away from the main circuit board **1541** so that a board surface of the branch circuit board **1543** may be perpendicular to a board surface of the main circuit board **1541** and the branch circuit board **1542**. Further, the second microphone **1532b** may be fixed at the peripheral side wall **1511** of the core housing in a direction facing away from the main circuit board **1541** and the branch circuit board **1542**.

In some embodiments, a pad **155**, a second pad (not shown in figures), the first microphone **1532a**, and the second microphone **1532b** may be disposed on the same side of the flexible circuit board **154**. The second pad may be disposed adjacent to the second microphone **1532b**.

In some embodiments, the second pad may be specifically disposed at one end of the branch circuit board **1543** away from the main circuit board **1541**, and have the same orientation as the second microphone **1532b** and disposed at intervals. Therefore, the second pad may be perpendicular to the orientation of the pad **155** as the branch circuit board **1543** is bent. It should be noted that the board surface of the branch circuit board **1543** may not be perpendicular to the board surface of the main circuit board **1541** after the branch circuit board **1543** is bent, which may be determined according to the arrangement between the peripheral side wall **1511** and the bottom end wall **1512**.

In some embodiments, another side of the flexible circuit board **154** may be disposed with a rigid support plate **4a** and a microphone rigid support plate **4b** for supporting the pad **155**. The microphone rigid support plate **4b** may include a rigid support plate **4b1** for supporting the first microphone **1532a** and a rigid support plate **4b2** for supporting the second pad and the second microphone **1532b** together.

In some embodiments, the rigid support plate **4a**, the rigid support plate **4b1**, and the rigid support plate **4b2** may be mainly used to support the corresponding pads and the microphone, and thus may need to have strengths. The materials of the three may be the same or different. The specific material may be polyimide (PI), or other materials that may provide the strengths, such as polycarbonate,

polyvinyl chloride, etc. In addition, the thicknesses of the three rigid support plates may be set according to the strengths of the rigid support plates and actual strengths required by the pad 155, the second pad, the first microphone 1532a, and the second microphone 1532b, and be not specifically limited herein.

The first microphone 1532a and the second microphone 1532b may correspond to two microphone components 4c, respectively. In some embodiments, the structures of the two microphone components 4c may be the same. A sound inlet 1513 may be disposed on the core housing. Further, the speaker may be further disposed with an annular blocking wall 1514 integrally formed on the inner surface of the core housing, and disposed at the periphery of the sound inlet 1513, thereby defining an accommodation space 1515 connected to the sound inlet 1513.

Referring to FIG. 15, FIG. 16, and FIG. 17, in some embodiments, the microphone component 4c may further include a waterproof membrane component 4c1.

As used herein, the waterproof membrane component 4c1 may be disposed inside the accommodation space 1515 and cover the sound inlet 1513. The microphone rigid support plate 4b may be disposed inside the accommodation space 1515 and located at one side of the waterproof membrane component 4c1 away from the sound inlet 1513. Therefore, the waterproof membrane component 4c1 may be pressed on the inner surface of the core housing. In some embodiments, the microphone rigid support plate 4b may be disposed with a sound inlet 4b3 corresponding to the sound inlet 1513. In some embodiments, the microphone may be disposed on one side of the microphone rigid support plate 4b away from the waterproof membrane component 4c1 and cover the sound inlet 4b3.

As used herein, the waterproof membrane component 4c1 may have functions of waterproofing and transmitting the sound, and closely attached to the inner surface of the core housing to prevent the liquid outside the core housing entering the core housing via the sound inlet 1513 and affect the performance of the microphone.

The axial directions of the sound inlet 4b3 and the sound inlet 1513 may overlap, or intersect at an angle according to actual requirements of the microphone, etc.

The microphone rigid support plate 4b may be disposed between the waterproof membrane component 4c1 and the microphone. On the one hand, the waterproof membrane component 4c1 may be pressed so that the waterproof membrane component 4c1 may be closely attached to the inner surface of the core housing. On the other hand, the microphone rigid support plate 4b may have a strength, thereby playing the role of supporting the microphone.

In some embodiments, the material of the microphone rigid support plate 4b may be polyimide (PI), or other materials capable of providing the strength, such as polycarbonate, polyvinyl chloride, or the like. In addition, the thickness of the microphone rigid support plate 4b may be set according to the strength of the microphone rigid support plate 4b and the actual strength required by the microphone, and be not specifically limited herein.

FIG. 18 is a partially enlarged view illustrating part C in FIG. 17 according to some embodiments of the present disclosure. As shown in FIG. 18, in some embodiments, the waterproof membrane component 4c1 may include a waterproof membrane body 4c11 and an annular rubber gasket 4c12. The annular rubber gasket 4c12 may be disposed at one side of the waterproof membrane body 4c11 towards the

microphone rigid support plate 4b, and further disposed on the periphery of the sound inlet 1513 and the sound inlet 4b3.

As used herein, the microphone rigid support plate 4b may be pressed against the annular rubber gasket 4c12. Therefore, the waterproof membrane component 4c1 and the microphone rigid support plate 4b may be adhered and fixed together.

In some embodiments, the annular rubber gasket 4c12 may be arranged to form a sealed chamber communicating with the microphone and only through the sound inlet 4b3 between the waterproof membrane body 4c11 and the rigid support plate. That is, there may be no gap in a connection between the waterproof membrane component 4c1 and the microphone rigid support plate 4b. Therefore, a space around the annular rubber gasket 4c12 between the waterproof membrane body 4c11 and the microphone rigid support plate 4b may be isolated from the sound inlet 4b3.

In some embodiments, the waterproof membrane body 4c11 may be a waterproof and sound-transmitting membrane and be equivalent to a human eardrum. When an external sound enters via the sound inlet 1513, the waterproof membrane body 4c11 may vibrate, thereby changing an air pressure in the sealed chamber and generating a sound in the microphone.

Further, since the waterproof membrane body 4c11 may change the air pressure in the sealed chamber during the vibration, the air pressure may need to be controlled within an appropriate range. If it is too large or too small, it may affect the sound quality. In the embodiment, a distance between the waterproof membrane body 4c11 and the rigid support plate may be 0.1-0.2 mm, specifically 0.1 mm, 0.15 mm, 0.2 mm, etc. Therefore, the change of the air pressure in the sealed chamber during the vibration of the waterproof film body 4c11 may be within the appropriate range, thereby improving the sound quality.

In some embodiments, the waterproof membrane component 4c1 may further include an annular rubber gasket 4c13 disposed on the waterproof membrane body 4c11 towards the inner surface side of the core housing and overlapping the annular rubber gasket 4c12.

In this way, the waterproof membrane component 4c1 may be closely attached to the inner surface of the core housing at the periphery of the sound inlet 1513, thereby reducing the loss of the sound entered via the sound inlet 1513, and improving a conversion rate of converting the sound into the vibration of the waterproof membrane body 4c11.

In some embodiments, the annular rubber gasket 4c12 and the annular rubber gasket 4c13 may be a double-sided tape, a sealant, etc., respectively.

In some embodiments, the sealant may be further coated on the peripheries of the annular blocking wall 1514 and the microphone to further improve the sealing, thereby improving the conversion rate of the sound and the sound quality.

In some embodiments, the flexible circuit board 154 may be disposed between the rigid support plate and the microphone. A sound inlet 1544 may be disposed at a position corresponding to the sound inlet 4b3 of the microphone rigid support plate 4b. Therefore, the vibration of the waterproof membrane body 4c11 generated by the external sound may pass through the sound inlet 1544, thereby further affecting the microphone.

Referring to FIG. 16, in some embodiments, the flexible circuit board 154 may further extend away from the microphone, so as to be connected to other functional components or wires to implement corresponding functions. Correspond-

29

ingly, the microphone rigid support plate **4b** may also extend out a distance with the flexible circuit board in a direction away from the microphone.

Correspondingly, the annular blocking wall **1514** may be disposed with a gap matching the shape of the flexible circuit board to allow the flexible circuit board to extend from the accommodation space **1515**. In addition, the gap may be further filled with the sealant to further improve the sealing.

It should be noted that the above description of the microphone waterproof is only a specific example, and should not be considered as the only feasible implementation. Obviously, for those skilled in the art, after understanding the basic principles of microphone waterproofing, it is possible to make various modifications and changes in the form and details of the specific method and step of implementing the microphone waterproof without departing from this principle, but these modifications and changes are still within the scope described above. For example, the count of the sound inlets **1513** may be set as one or multiple. All such modifications are within the protection scope of the present disclosure.

It's noticeable that above statements are preferable embodiments and technical principles thereof. A person having ordinary skill in the art is easy to understand that this disclosure is not limited to the specific embodiments stated, and a person having ordinary skill in the art can make various obvious variations, adjustments, and substitutes within the protected scope of this disclosure. Therefore, although above embodiments state this disclosure in detail, this disclosure is not limited to the embodiments, and there can be many other equivalent embodiments within the scope of the present disclosure, and the protected scope of this disclosure is determined by following claims.

What is claimed is:

1. A speaker, comprising:

a core housing configured to accommodate an earphone core;

a transducer residing inside the core housing and configured to generate vibrations, the vibrations producing a sound wave inside the core housing and causing a leaked sound wave spreading outside the core housing; at least one sound guiding hole located on the core housing and configured to guide the sound wave inside the core housing through the at least one sound guiding hole to an outside of the core housing, the guided sound wave having a phase different from a phase of the leaked sound wave, the guided sound wave interfering with the leaked sound wave in a target region;

a first microphone and a second microphone disposed in the core housing, wherein the first microphone and the second microphone are disposed at different positions of a circuit board; and

a first sound inlet and a second sound inlet disposed on the core housing, wherein the first microphone and the second microphone collect external sound through the first sound inlet and the second sound inlet, respectively, the first sound inlet and the second sound inlet are set to face different directions, and a main axis of a sound receiving region of the first microphone is inclined relative to a main axis of a sound receiving region of the second microphone,

wherein the circuit board includes a main circuit board, and a first branch circuit board and a second branch circuit board connected to the main circuit board, the first microphone is disposed on one end of the first branch circuit board away from the main circuit board, the second microphone is disposed on one end of the

30

second branch circuit board away from the main circuit board, and a portion of the second branch circuit board on which the second microphone is disposed is bent relative to the main circuit board so that a board surface of the second branch circuit board is inclined relative to a board surface of the first branch circuit board.

2. The speaker of claim 1, wherein the first branch circuit board extends in a same direction as the main circuit board, the second branch circuit board extends perpendicular to the main circuit board.

3. The speaker of claim 1, wherein a plurality of pads are disposed on an end of the main circuit board away from the first branch circuit board and the second branch circuit board.

4. The speaker of claim 1, wherein the first microphone and the second microphone are disposed on a same side of the circuit board.

5. The speaker of claim 1, wherein the core housing includes a peripheral side wall and a bottom end wall connected to one end surface of the peripheral side wall to form an accommodation space with an open end.

6. The speaker of claim 5, wherein the first sound inlet is disposed on the peripheral side wall, and the second sound inlet is disposed on the bottom end wall.

7. The speaker of claim 6, wherein the first microphone is disposed on the peripheral side wall, and the second microphone is disposed on the bottom end wall.

8. The speaker of claim 1, further comprising a waterproof membrane component that is attached to an inner surface of the core housing at a periphery of the first sound inlet.

9. The speaker of claim 8, further comprising:

a rigid support plate pressed the waterproof membrane component on the inner surface of the core housing, the rigid support plate being disposed with a third sound inlet.

10. The speaker of claim 9, wherein the waterproof membrane component includes a waterproof membrane body and a first annular rubber gasket disposed on one side of the waterproof membrane body towards the rigid support plate, wherein

the first annular rubber gasket is disposed on the periphery of the first sound inlet and a periphery of the third sound inlet; and

the rigid support plate is pressed against the first annular rubber gasket.

11. The speaker of claim 10, wherein the first annular rubber gasket is arranged to form a sealed chamber communicating with the first microphone only through the third sound inlet of the rigid support plate.

12. The speaker of claim 10, wherein the waterproof membrane component further includes a second annular rubber gasket disposed on one side of the waterproof membrane body towards the inner surface of the core housing.

13. The speaker of claim 9, wherein the core housing includes an annular blocking wall protruding inwardly relative to the inner surface of the core housing and disposed on the periphery of the first sound inlet, thereby defining an accommodation space connected to the first sound inlet.

14. The speaker of claim 13, wherein the waterproof membrane component and the rigid support plate are disposed in the accommodation space.

15. The speaker of claim 1, wherein the main axis of the sound receiving region of the first microphone is perpendicular to the main axis of the sound receiving region of the second microphone.

31

16. The speaker of claim **1**, wherein the first microphone is arranged facing to the first sound inlet, and the second microphone is arranged facing to the second sound inlet.

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32